Gap position, not gap length, governs the error of annual thermal extremes in incomplete river-temperature records

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Key points

- A hottest 30-day mean grows 9.6-fold along gap position and 3.5-fold along gap length, while an annual mean reverses the two
- Support-conditioned screening cuts pooled designed-outage errors by 7.1 points; the gaps these networks actually keep give only 0.7 to 1.9
- On 2422 real incomplete years the two admission rules disagree on 15.1%, and the count of wholly hidden peaks is a proxy, not truth

Abstract

Thermal archives are screened by completeness thresholds, assuming that how much data is missing decides whether an annual statistic can be trusted. That assumption has never been tested against the statistic's own support, because a gap's length and its position cannot be varied independently in a record that failed on its own. Here we combine 5114 complete river station-years from 146 networks with 51524 designed outages, vary the two properties of a gap separately, and score admission rules on held-out networks and on sealed panels that took no part in development. We find that the error of a hottest 30-day mean grows 9.6× along the position axis and 3.5× along the length axis, whereas an annual mean reverses the two (1.5× and 10.2×): the same gap is destructive or harmless depending only on which statistic is being computed. Screening station-years by the share of that statistic's support a gap hides cuts consequential errors on a retrospective network holdout from 9.7% to 2.5% at 70% retention (-7.1 points, $[-9.0, -5.4]$), by -7.9 points ($[-10.2, -5.8]$) on 39 sealed USGS networks and by -4.6 ($[-6.5, -3.3]$) on a pre-registered multi-agency panel of 32 networks, yet by only -0.7 to -1.9 points on the gaps those same networks actually experience. Completeness thresholds are therefore the wrong axis for extremes and the right one for means; what an agency can bank today is the smaller empirical-arm gain, and that difference is what its own gap calendar decides.

Plain Language Summary

Rivers are getting warmer, and managers ask how hot a river got last summer. The answer needs a year of daily temperatures, but records have holes, and the usual screen counts how many days are missing. We show that this is the wrong question for a summer peak: what matters is whether the missing days include the few weeks the peak depends on. Using complete years from 146 river networks, we place gaps of identical length in different positions and compare what two statistics lose. A gap over the peak window destroys the hottest 30-day mean while the same gap in autumn barely moves it; a whole-year average behaves in the opposite way. A rule that screens on this "hidden support" instead of on gap length removes most large errors on held-out and sealed networks, but when we look at the gaps these networks really experience the gain shrinks to a fraction. Results about exact crossover days, autumn maintenance benefits and "peak never observed" counts are exploratory or use proxies and should not yet guide operations.

1 Introduction

Cold-water habitat assessment, thermal-regime classification and criterion-exceedance accounting all reduce a year of daily water temperature to one number — a hottest 30-day mean, an annual mean, a seasonal amplitude (Caissie, 2006; Isaak et al., 2017) — and it is that number, not the daily series, that enters a management judgement. The error that matters is therefore the error of the derived statistic. On the panel assembled here, computing the summer peak from recorded days alone, which is the default once a station-year passes a completeness screen, carries a mean absolute error of **1.305 °C**, or 1.23 year-to-year standard deviations of the quantity itself, and 24.9% of station-years are wrong by more than 1.0 °C. An error of that size turns a thermally stressful year into an unremarkable one.

Current practice assumes that the amount of missing data decides whether the derived number may be used, and two families of rule encode that assumption (Table 1). Collection standards specify what must be present: Mauger et al. (2015), writing minimum standards for a state whose records feed regional analyses, require a minimum sampling duration of one calendar month and recommend a deployment period of 1 June to 31 August "to capture the thermal maxima"; the same standards govern which of those records enter the regional assessment (Mauger et al., 2017). Quantitative guidance constrains what may be absent: Johnson et al. (2021) report that "datasets with less than seven weeks of consecutively missing data enabled estimation of all annual signal parameters with reasonable accuracy", expanding to "approximately 20 weeks" when missing data are imputed. The gap-filling literature optimises the accuracy of the filled series rather than the admissibility of the completed year (Mohseni et al., 1998; Li et al., 2017), and the closest prior work fills incomplete summer records so that a thermal peak can be computed, applies that filling uniformly, declines no station-year, and admits July–August-only records on the climatological position of the hottest window (Beaufort et al., 2022). Large-sample thermal models optimise daily or reach-scale prediction skill (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023), and selective-prediction methods control error at a fixed rejection rate (Geifman & El-Yaniv, 2017; Zaoui et al., 2020). None of these rules measures how much of the target statistic's own support a record contains.

The assumption is plausible for some statistics and wrong for others. An annual mean is a sum over days, so removing k days removes k terms and it matters little which. A hottest 30-day mean is an extreme supported on a short window: if the outage misses that window the statistic is recoverable almost exactly however long the outage is, and if the outage covers it the statistic must be inferred however short the outage is. The direction of a completeness rule's error is therefore opposite in the two cases, and an average over a mixed archive conceals the reversal. The literature is consistent with that reversal without having isolated it: reported timing effects on annual thermal parameters are inconsistent across regions, years and parameters (Johnson et al., 2021); reconstruction bias depends on where the target window falls (Beaufort et al., 2022); systematic, time-varying instrument bias rather than random noise dominates long-term monitoring error (Jackson et al., 2025); and statistic-aware completeness conventions do exist in climate monitoring, but they are defined by counts or seasons rather than by the statistic's support (Zhang et al., 2011). Irregular sampling and sampling artifacts likewise bias means and trends and decide which model structures remain trustworthy (Grey et al., 2024), and reconstruction frameworks have been built for regulated and data-sparse basins, including Chinese catchments and the Yangtze mainstem (Huang et al., 2023; Wang, 2025); those studies answer estimation and reconstruction questions, not the archive-admission question for a named annual extreme.

Testing the assumption requires separating two properties of a gap that real records entangle. Gap length and the share of the target window that is hidden correlate at -0.86 in our own panel, so a comparison across real gaps cannot say whether a rule works because it avoids long gaps, because it avoids badly placed ones, or because both simply track how much record the station has. A second obstacle is truth: in exactly the years where the admission decision matters, the target statistic is unknown by construction, so the benefit of a rule cannot be measured where it would be applied. A third is that evaluations report a rule's discrimination rather than the composition of the archive it leaves behind, and composition is what decides which catchments remain available for analysis.

To address these limitations we assemble 5114 complete river station-years from 146 networks and nine national agencies, place 51524 outages on complete years so that the two properties of a gap can be varied independently, and score

admission rules on held-out river networks, on sealed panels that took no part in development, and — without any truth — on the 2422 records in this same archive that really do have gaps. Three questions organise the paper, and the last matters most, because a screening rule is judged by the archive it leaves behind. First, which property of a gap governs the error of a derived thermal statistic, and does the answer depend on which statistic is being computed? Second, is that dependence a dichotomy between two regimes or a continuous transition, where does it turn over, and how much of the gain survives the gap calendar these networks actually keep? Third, what does conditioning on it change for the error among retained station-years, for which stations survive in the archive, and for the undecided records?

2 Study system, data and identification design

2.1 The statistic, its support, and the share a gap hides

Let a station-year be a sequence $y = (y_1, \dots, y_T)$ of daily mean water temperatures, $T \in \{365, 366\}$, and let $q(\cdot)$ be the derived statistic of interest. The family we sweep is the hottest consecutive k-day mean

$$q_k(y) \;=\; \max_{t} \; \frac{1}{k} \sum_{i=t}^{t+k-1} y_i, \tag{1}$$

which is the annual maximum at $k = 1$ and the annual mean at $k = T$; the coldest 30-day mean, the annual amplitude, and the sine-fit mean, phase and amplitude of Johnson et al. (2021) are scored alongside.

A gap is a binary mask $m \in \{0,1\}^T$; the analyst observes the days where $m_t = 0$. A reconstruction $\hat{y} = R(y_{\bar{m}}, Z)$ completes the year using side information Z — synchronous donor stations, air temperature, the station's own history — and reports $q(\hat{y})$. The quantity that matters is the error of the reported statistic,

$$L \;=\; \big|q(\hat{y}) - q(y)\big|, \tag{2}$$

and a station-year is called *consequential* when $L > \tau$ for a tolerance τ in the units of q.

The property of the gap we propose as the governing one follows from how much each day moves the statistic. Writing $w_t^q = |\partial q(y)/\partial y_t|$ for the discrete influence function of q at y (Hampel, 1974), the *hidden mass* of a gap is the share of that

Table 1. What each body of work optimises, decides on, and conditions on. The rows are ordered by what they assume about the conditioning variable, not by date.

body of work	optimises	decision unit	conditioning axis
completeness guidance (Johnson et al., 2021)	tolerable amount of missing data	station-year	outage size (timing tested; effects reported inconsistent)
collection standards (Mauger et al., 2015)	comparability across a programme	deployment	season and duration
gap filling (Mohseni et al., 1998; Li et al., 2017)	accuracy of the filled series	day	air temperature, donors
peak reconstruction (Beaufort et al., 2022)	spatial coverage of the thermal peak	reach	air regression applied uniformly; climatological summer window used to admit partial-season records
large-sample models (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023)	prediction skill	day or reach	meteorology, catchment attributes
selective prediction (Geifman & El-Yaniv, 2017; Zaoui et al., 2020)	error at a fixed rejection rate	case	model confidence
this work	**error of the derived statistic among retained years**	**station-year**	**share of the statistic's support hidden**

influence it removes,

$$h_q(m) \;=\; \frac{\sum_{t:\, m_t=1} w_t^q}{\sum_t w_t^q}. \tag{3}$$

For the annual mean every w_t is equal and h_q is the missing fraction; for the hottest k-day mean the influence is concentrated on the k days achieving the maximum and h_q is the share of that window the gap hides; for the sine-fit parameters the weights are smooth functions of day of year — a constant for the fitted mean, $|\sin(\theta_t - \phi)|$ for the amplitude, which is largest at both extremes, and $|\cos(\theta_t - \phi)|$ for the phase, which is largest a quarter-year from the peak. Equation (3) therefore contains the incumbent axis as a special case rather than opposing it, which is why the two regimes reported below are ends of one scale.

Equation (3) cannot be evaluated at decision time, because the weights depend on the days the gap hides. What an analyst has is the *climatological* support: the window the station's other complete years say the maximum should fall in. That estimate has an error, and Section 4.3.1 shows the error is large enough to change what a screening rule should condition on.

2.2 Records, and what each dataset is for

Daily water-temperature series were assembled from nine national providers: ARSO (Slovenia), CHMI (Czechia), FOEN (Switzerland), GKD Bayern and LUBW (Germany), NVE (Norway), Rijkswaterstaat (Netherlands), SYKE (Finland) and the USGS (United States). Acquisition contracts differ, and we audited each one against the provider's own documentation (`measurement_provenance_dictionary.csv`). Five providers document a daily mean and the parser reads it: USGS (`parameterCd=00010`, `statCd=00003`), FOEN (`data_1day_mean`), GKD Bayern (`Tageswerte`, column `Mittelwert`), NVE HydAPI (resolution 1440, documented as mean) and, for the transport panel below, the UK Environment Agency (measure period 86400, "daily mean Temperature"), with LUBW's `Tagesmittelwert` documented but its opaque workbook column only partially re-verified. Four providers do not meet that contract as read: CHMI's parser averages 10-minute *instantaneous* values (T_O) even though the provider also publishes a daily-mean product (T_D); Rijkswaterstaat's parser averages raw `NVT` readings even though a live 24 h mean parameter (`GEM24H`) exists; SYKE's public day values are morning nearshore reference readings

(08:00, 20 cm below the surface) rather than 24 h means; and ARSO's daily value has no published mean-versus-hour statement. We therefore treat "daily mean" as a contract that holds for part of the panel and report the consequence: restricted to the five providers whose daily value is a documented provider daily mean, the development contrast is −5.7 points ([−6.5, −4.9], 84 networks) and the reserved contrast −6.6 ([−7.9, −5.3], 29 networks), against −6.4 and −7.1 for the full panel (`amendments/provider_contract_sensitivity/`). The sealed confirmation is USGS-only and so is entirely inside the documented stratum. The three SYKE stations remain flagged in `results/wrr_submission_ready/measurement_provenance_stations.csv` and are read as a sensitivity stratum, not as proven daily means. A *network* is one provider's group of stations on a common river system. It is the unit at which donor stations are available to one another, and it is the clustering unit for every interval reported here, because station-years within a network share weather, regulation and instrumentation.

Waterbody class was assigned per network and non-river objects — seas, lakes, canals, estuaries, deltas — are excluded rather than averaged into a river result. Reference series are never imputed: a calendar year with any missing day supplies no truth. Stations with fewer than four fully observed calendar years are excluded because the estimator protocol requires them. This leaves **146 networks, 457 stations and 5114 fully observed station-years**.

The same 457 stations contribute **2422 further station-years that are genuinely incomplete**. These carry no truth and are therefore not part of the scored population; they are the population a screening rule exists for. Section 4.4.3 reports overlap-rule decisions on that panel; the frozen certificate can also be applied through a truth-free `predict` interface (repository amendment `results/wrr_submission_ready/e2_certificate_predict/`).

Station coordinates were assembled for all 457 stations, from the providers' own catalogues, from the USGS NWIS site service for 195 United States gauges, and from ARSO's public station feed, and daily 2 m air temperature was joined from NASA POWER for the whole panel, 1981 to 2026 (NASA Langley Research Center, 2026). For the United States subset, basin characteristics were joined from GAGES-II (Falcone, 2011) on the eight-digit gauge identifier, matching **140 of 204**

eligible stations across 57 networks. Basin attributes are used to ask which rivers are hard (Section 4.4.5) and which rivers a screen keeps (Section 4.4.4); they are deliberately not used inside the screening rule, because they are unavailable for the other eight agencies.

Table 2 lists every quantity the analysis uses, what it means physically, where its value comes from, and whether it was calibrated. Networks — never station-years — were split into development and reserved, stratified on provider and on donor regime, with the seed recorded before any scoring. This gives **110 development and 36 reserved networks**. That reserved panel is a retrospective network hold-out for the frozen v17 screen: 28 of 36 networks have prior development, descriptive or scoring use in earlier programme stages (`results/wrr_submission_ready/prior_access_ledger_reserved_networks.csv`). It is not an unused confirmation set for analyses developed after opening. A sealed confirmation was therefore recruited externally: 39 USGS HUC4 networks (91 stations; 18 HUC2 contexts; 8973 designed outage cases) that lie outside the v17/v18 development panel were ingested, hashed, and scored once with the frozen certificate (`p0_07_sealed_confirmation/verdict.json`). At 70% retention and 0.5 °C the pooled contrast is about −7.9 percentage points (clustered interval about $[-10.2, -5.8]$); the empirical arm is smaller but excludes zero (about −1.7 points, $[-3.0, -0.7]$). The panel is outside the development programme but it is not untouched by the wider programme, and we report that rather than assert otherwise: a repo-wide, provider-aware audit of every artifact carrying a station identifier finds 24 of the 39 networks appearing in earlier iterations, 15 networks with no outcome-bearing prior use and 4 with no prior use of any kind (`prior_access_audit_sealed_usgs.csv`; `AMENDMENT_prior_access_sealed_panel.md`). The result does not depend on the touched networks: restricted to the 15 outcome-clean networks the pooled contrast is −9.4 points ($[-12.4, -6.0]$) with an empirical-arm gain of −3.7 ($[-5.8, -1.6]$), and restricted to the 45 outcome-clean stations it is −8.3 ($[-10.4, -6.4]$). Those subsets are read from an already-sealed score and are a transparency diagnostic, not new confirmation. The panel remains USGS-only and uses provisional HUC4 groupings rather than agency river systems, so a pre-registered transport test was recruited to ask whether the result travels. Panels from Sweden (SMHI), England (UK Environment Agency) and France (Hub'Eau) — 32 networks, 245 stations, 17 826 designed outage cases

across boreal, temperate-oceanic and temperate-continental rivers, none of them used anywhere in the development programme — were sealed with input hashes and scored once with the same frozen configuration, against a rule fixed before unsealing ($\Delta \leq -1$ point with a clustered interval excluding zero and at least two providers in the same direction); twenty-one additional Norwegian basins enter only as a labelled secondary stratum. The protocol records that this test was designed after the USGS score was known, so it is a transport test and not a pristine first confirmation. Section 2.2 reports what it found.

2.3 Gap design: two arms, and why both are needed

A case is one station-year under one gap realization. Two arms are used and reported separately, because they answer different questions and are not one population. The *empirical* arm samples real gap runs — length and start month — from the same network's own records and transplants them onto fully observed years, so that operational numbers reflect the geometry agencies experience. The *stratified* arm crosses gap length $\{14, 30, 60, 90, 120, 180, 270\}$ days with the overlap between the gap and the station's climatological peak window, so that the response surface is estimable rather than accidental. The design yields **51524 cases**, 40704 development and 10820 reserved.

Both arms transplant a mask onto a year whose values were observed, which assumes the mask is uninformative about the values it hides. Section 4.4.6 tests that assumption against the archive rather than asserting it. Gaps are placed inside one calendar year; failures that straddle New Year damage two annual statistics while hiding neither warm season, and the design cannot represent that case.

Each endpoint is scored against a tolerance equal to the same fraction, 0.47, of *its own* interannual standard deviation at the same stations. For the annual peak that scale is 1.06 °C, giving a primary tolerance of 0.50 °C; the scale falls monotonically with support width, from 1.18 °C at $k = 1$ to 0.55 °C at $k = 365$. The fraction is swept at 0.25, 0.47 and 1.0, and no conclusion about which axis wins is claimed unless it is invariant across the sweep. The reference series carries instrument error of roughly 0.1 to 0.2 °C, so the primary tolerance is about three times sensor accuracy; Section 4.4.6 perturbs the reference by 0.15 °C and repeats the analysis.



Figure 1. Study design and data flow. Nine national providers are ingested and quality-checked into 146 river-only networks, 457 stations and 5114 complete station-years; 51 524 designed outages are placed on those complete years and split by network into a 110-network development set and a 36-network reserved panel, and the configuration is frozen. The reserved panel is a retrospective holdout with mixed prior access; the sealed USGS confirmation (39 networks, 8973 cases, with a repository-wide prior-access audit) and the pre-registered multi-agency transport panel (32 primary networks across three contexts, plus a 21-network secondary stratum) were recruited afterwards and scored once each. The same frozen configuration is then applied, without any truth, to the 2422 real incomplete station-years. Source: draw.io source `figures/v17/drawio/f2_experiment_design.drawio`.

Table 2. Quantities used in the analysis.

quantity	physical meaning	source	calibrated
h_q, hidden mass	share of the statistic's influence removed by the gap	mask and the station's own complete years	no
peak-window overlap	h_q for a 30-day extreme, from the climatological window	mask and training years	no
warm-season coverage	share of May–September days observed	mask	no
longest run, missing fraction	size of the gap	mask	no
observed-peak margin	how far the observed days already constrain the annual maximum	observed days and training climatology	no
donor count, coverage, correlation	how much synchronous anomaly information neighbours carry	training-period deseasonalised series	no
K, ridge penalty of the donor and air estimators	strength of shrinkage toward the seasonal mean	leave-one-year-out within station	yes, within station
conformal scale	multiplicative factor turning a predicted bound into a valid one	calibration networks, disjoint from fitting and test	yes, on networks
τ, tolerance	consequential error in the units of the statistic	0.47 of the endpoint's own interannual SD; swept 0.25–1.0	no, declared
basin attributes	elevation, slope, baseflow index, drainage area	GAGES-II, United States subset	no, not an input

aNo quantity in the first six rows reads a day the gap hides; that restriction is enforced in the feature constructor rather than by convention.

3 Methods

Each choice below answers one of the obstacles in the introduction, and the order of the subsections follows that logic rather than the order in which the code runs: the reconstruction defines what a deployed rule may read; the rule defines the two candidate conditioning axes and the information barrier between them and the truth; the test protocol fixes the operating point, the clustered inference unit and the falsification tests declared before scoring.

3.1 Reconstruction, and what each estimator is allowed to see

Every case is completed by each supported estimator: the statistic computed from recorded days alone (current practice), a day-of-year median climatology, a smoothed climatology, linear interpolation where both flanks exist, ridge regression on the visible subset of synchronous donors, and an air-driven ridge with lagged air temperature and harmonics. Estimators are fit leave-one-year-out within the station, which is what offline archive repair permits; Section 4.4.6 repeats the study under the stricter protocol in which only prior years are available. Support is declared per method: donor regression does not read a flanking observation, so requiring one would discard usable cases, and absent donor columns are never imputed but the model is refit on the visible subset.

One weakness in that library was worth removing before the comparison was run. Air temperature was available for 2.3% of development cases in an earlier version of this analysis, so the reconstruction this field actually uses was effectively absent, and 36 of the 110 development networks have no synchronous donor at all. With coordinates joined for all 457 stations and NASA POWER pulled for the whole panel, air forcing now covers 98.6% of development cases and 100% of reserved cases. The retrospective reserved holdout is not re-scored against the new forcing; Section 4.4.2 reports it as a development-panel sensitivity analysis.

The reconstruction layer is not the contribution of this paper and is kept deliberately simple. Its role is to make the comparison between conditioning variables fair, which requires only that the estimator paired with the incumbent rule is the strongest deployable one; Section 4.4.2 tests that requirement directly rather than assuming it.

3.2 The screening rule, and the information it is forbidden to read

For each case the screen predicts an upper bound on the absolute error each supported estimator would produce, selects the estimator with the smallest bound, and emits a conformal interval on the statistic together with an admission decision at an analyst-chosen tolerance. Its inputs are the first six rows of Table 2. No input may read a day the gap hides, and that restriction is enforced in code rather than by convention.

Calibration is split conformal (Lei et al., 2018), and the risk-control framing follows the distribution-free line of Angelopoulos et al. (2022) with the caveat that selective observation is itself an existing problem statement (Bai et al., 2026); what is specific here is that selection happens among reconstructions chosen per station-year. Two properties matter. It is *selection-conditional*, because taking the smallest of several bounds is a winner's curse that per-estimator calibration cannot see, and *multiplicative*, because the error scale spans an order of magnitude between a gap that misses the peak window and one that covers it. Fitting, calibration and test networks are always disjoint.

3.3 Test protocol: fixed operating point, clustered unit, declared falsifications

Before any rule was fit we measured, on development cases, how much a rule with perfect foresight could win inside each candidate action space, and compared it with the margin the reserved panel can resolve. Two natural framings — adopt or reject a fixed reconstruction, and choose among reconstruction methods — have ceilings of $+0.038$ and $+0.080\,^{\circ}\mathrm{C}$ against a threshold of $0.136\,^{\circ}\mathrm{C}$, and were not pursued. Retain-or-exclude clears its gate by a factor of five and is the action space studied here. A decision experiment whose ceiling is below its own resolvable margin cannot produce evidence in either direction, and running one consumes a reserved panel for nothing.

Every interval resamples whole networks, because station-years within a network are not independent and reusing a station-year under several gap realizations does not create independent observations. The pre-registered primary criterion, frozen before the reserved panel was opened, is that at the retention of the matched

length rule the screen must reduce the consequential-error rate by at least five percentage points with a network-clustered interval excluding zero. Precision was computed in advance: the reserved panel resolves 1.44 points, so the declared effect is 3.5 times the resolution.

Where several endpoints are compared, the direction of each contrast was predicted in advance from the form of the statistic, which is the defence against multiplicity. Three of those predictions were not met, and Sections 4.3.1 and 4.2.2 report them as failures rather than reshaping them into successes.

4 Results

4.1 The phenomenon: position, not length, decides what an extreme loses

Whether an incomplete year can carry a summer extreme depends on where its gap falls and hardly at all on how long the gap is. One real record shows the two ends of the range: a 30-day gap over the peak window costs 3.99 °C in the hottest 30-day mean, the same 30 days in late autumn costs nothing, and the annual mean responds in the opposite order. The reversal is systematic across five annual endpoints, and it is large enough to change which station-years an archive should keep.

4.1.1 Computing the peak from recorded days alone costs more than a year-to-year standard deviation

Current practice is expensive before any rule is proposed. On development cases, computing the annual peak from recorded days alone — the default once a station-year passes a completeness screen — gives a mean absolute error of **1.305 °C, or 1.23 interannual standard deviations** of the quantity being measured, with 24.9% of station-years wrong by more than 1.0 °C. Substituting a day-of-year climatology over the gap gives 0.402 °C, the best supported reconstruction 0.225 °C with air forcing available for the whole panel (0.242 °C without it), and an oracle over the estimator set 0.130 °C. The error is therefore dominated not by which station-years are kept but by whether the record is repaired at all, which fixes

the size of the question the rest of the paper addresses: given that a year has been repaired, which repaired years can be trusted.

4.1.2 The share of the support a gap hides orders every endpoint, and the ordering reverses

Table 3 reports, for five endpoints, how well each candidate property of the gap discriminates a consequential error. The ordering reverses between extremes and averages, every clustered interval excludes zero, and the winning axis is unchanged at tolerances of 0.25, 0.47 and 1.0 interannual standard deviations. The seven-day peak has the largest timing advantage of any endpoint, which the support account predicted before it was measured because a shorter extreme is more localised in time; peak-window overlap has an AUC of 0.447 for the cold extreme, because hiding the summer peak carries no information about the winter minimum. The last row is not evidence that cold extremes are a length-governed exception, and reading it that way would be a mistake: the two support axes in the table are defined on the warm season. Built on the endpoint's own season — a climatological coldest-30-day window and the observed share of the cold months — the support axis beats the length rule for the coldest 30-day mean on the same development cases (AUC 0.876 and 0.897 against 0.819; clustered error-rate gaps -0.0050 and -0.0059, intervals excluding zero), and adds to length in a grouped out-of-fold model (AUC 0.905 against 0.813 for length alone). What the table shows is that the support must be defined on the season the statistic lives in; it does not license a warm-season axis for a cold endpoint. That cold-endpoint analysis is exploratory: it uses the development panel and does not touch the sealed confirmation.

4.2 The response is conditional: where the gain holds and where it disappears

The advantage of conditioning on support is not general. It depends on how wide the estimand's support is, it survives only in part under the gap calendar these networks actually keep, and it differs across agencies and across how strictly their daily values are checked. Reporting only the pooled number would hide all three dependencies.

Table 3. Discrimination of a consequential error by axis (AUC), development cases, each endpoint at 0.47 of its own interannual standard deviation.

endpoint	peak-window overlap	warm-season coverage	outage length	AUC difference (95% CI)
hottest 7-day mean	0.823	0.824	0.752	+0.072 (0.063, 0.096)
hottest 30-day mean	0.819	0.820	0.750	+0.070 (0.061, 0.098)
amplitude	0.758	0.792	0.768	+0.024 (0.015, 0.037)
annual mean	0.672	0.818	0.845	−0.027 (−0.036, −0.018)
coldest 30-day mean	0.447	0.619	0.803	−0.184 (−0.228, −0.156)

aThe difference is AUC(best timing axis) minus AUC(outage length). All five intervals exclude zero and the ordering reverses between the two regimes.

4.2.1 The transition is continuous, and a fixed narrow window turns over near a three-month support

Table 3 evaluates the question at two points and reads the gap between them as a dichotomy. Sweeping the hottest-k-day mean of Equation (1) from $k = 1$ to $k = 365$ evaluates it everywhere (Table 4, Figure 4a). The advantage of a fixed 30-day peak window — the axis the screen actually uses — is flat at about +0.075 AUC while the estimand's support is a month or less, falls to +0.050 at two months and to +0.002 at three, and is negative beyond that. The 90-day grid point is statistically indistinguishable from zero (95% interval −0.018 to +0.026, positive in 55% of network-clustered replicates), and the first clearly negative grid point is 182 days, so the crossing is only located between 60 and 182 days: a bootstrap of the whole curve puts its 95% interval at $[77, 107]$ days and it moves between 85 and 102 days across the tolerance grid, while the opened reserved panel's interval ($[104, 136]$) overlaps it rather than displacing it. We therefore report a range and not a day count. The same sweep supports a second, different statement: a conditioning win-

Table 4. The governing axis as a function of the support of the estimand, development cases.

support width k (days)	interannual SD (°C)	AUC, peak window	AUC, missing fraction	difference (95% CI)
1	1.18	0.825	0.751	+0.074 (0.053, 0.097)
3	1.17	0.824	0.746	+0.078 (0.055, 0.104)
7	1.16	0.823	0.746	+0.077 (0.053, 0.104)
15	1.12	0.821	0.743	+0.078 (0.053, 0.106)
30	1.08	0.819	0.744	+0.075 (0.049, 0.104)
60	1.02	0.815	0.765	+0.050 (0.028, 0.078)
90	0.94	0.792	0.790	+0.002 (−0.018, 0.026)
182	0.74	0.728	0.823	−0.095 (−0.111, −0.077)
365	0.55	0.672	0.845	−0.173 (−0.192, −0.157)

aThe difference is AUC(30-day peak window) minus AUC(whole-year missing fraction), with a network-clustered 95% interval. Every interval excludes zero except the one at $k = 90$, which is where the crossing is.

dow dilated around the estimand's own support keeps its advantage over whole-year completeness out to about 359 days (95% interval 351 to 364, and it fails to cross at all in 7% of replicates), ending only at the trivial $k = 365$ coincidence, and the advantage is a hump that peaks near 60 days rather than decaying monotonically. The mid-range crossing is thus a property of holding the conditioning window narrow and fixed, not of the support idea itself. Nor is one window enough across the continuum: a single 60-to-90-day window is within 0.0004 AUC of each endpoint's own optimum up to a 90-day support, but costs 0.083 AUC at the annual-mean end. These continuum numbers are exploratory analyses developed after the reserved panel was opened; they motivate the screening axis but are not the sealed USGS confirmation reported with the data description. The two regimes of Table 3 are useful ends of that exploratory scale, not a proven universal dichotomy.

4.2.2 On the estimands the standing guidance was derived for, the guidance is right

The sine-fit parameters also connect to a physical reading that is worth keeping separate from the statistical one: air–water phase and amplitude carry information about groundwater and riparian influence (Hare et al., 2023), and a support-conditioned screen improves the error of those parameters without claiming to identify the process behind them.

Johnson et al. (2021) fit a sine wave by linear regression and take a mean, a phase and an amplitude; they do not compute a hottest 30-day mean. Reproducing that estimator on our panel puts the two accounts on the same footing for the first time (Table 5, Figure 4c). A completeness threshold is the correct axis for the fitted mean, at 0.845 against 0.672 for peak-window overlap, and is within 0.02 AUC of the best axis for the phase. The ordering is the same under the authors' own protocol, in which the sine is fitted to whatever days are present with no repair at all: peak overlap 0.667, 0.743 and 0.590 against whole-year missingness 0.902, 0.842 and 0.898 for the mean, amplitude and phase, at base rates of 32% to 34% consequential against 10% to 12% after repair.

Two reporting rules come out of this endpoint. Phase is only identifiable when the fitted amplitude stands above the noise: the standard error of a three-parameter sine fit is $\sigma\sqrt{2/n}/A$ radians, so requiring a 95% interval no wider than the phase tolerance used here gives $A/\sigma \geq 4.5$, which retains 70.7% of cases and leaves the comparison above unchanged (+0.0126 against +0.0127 on all cases); under the authors' own unrepaired protocol the defect is active rather than latent, with 0.10% of cases below $A/\sigma = 1$ carrying a median phase error of 60 days against 91 days for a random phase. The rule is an identifiability floor, not a quality guarantee: the error rises again above $A/\sigma \approx 8$. Second, the fitted mean and phase are the regime the standing guidance was derived for, and on them it is right.

The third declared prediction failed here. We expected that for the phase, whose influence is largest a quarter-year from the peak, two 61-day windows around the spring and autumn limbs would beat both peak overlap and gap length. They beat peak overlap decisively (+0.249 AUC, 95% CI 0.231 to 0.270) and they do not beat gap length (0.831 against 0.836). Weighting every day by the influence function

Table 5. Axis comparison on the sine-fit parameters (AUC), development cases, each at 0.47 of its own interannual standard deviation.

estimand	peak window	missing fraction	outage length	influence-weighted	best axis
fitted mean	0.672	0.845	0.845	0.845	outage length
amplitude	0.724	0.812	0.815	0.831	influence-weighted
phase	0.583	0.835	0.836	0.847	influence-weighted

of Equation (3) instead recovers the advantage: 0.847, which beats the whole-year missing fraction by +0.013 (0.009, 0.017) and peak overlap by +0.265 (0.249, 0.285).

4.2.3 The gain disappears under the gap calendar these networks keep, because long gaps begin in autumn

The frozen holdout test was run on the pooled case set, which mixes two arms whose base rates differ by an order of magnitude, and they must be reported apart (Table 6). The declared five-point effect is met on the pooled and stratified populations and *not* on the reserved empirical arm, where the interval includes zero. The holdout headline −7.13 points is therefore an effect on a mixture that includes a designed arm. The sealed USGS confirmation likewise finds a much smaller empirical-arm gain than the pooled contrast, so neither number should be quoted as next year's agency experience under natural outages alone.

Why the empirical arm is null is itself a result. Across the 6472 real interior gaps in this archive, gaps shorter than 30 days and gaps of 30 to 89 days are close to uniform in start month ($\chi^2 = 4.0$ on 11 df, $p = 0.97$ for the 30-to-89-day class). Gaps of 90 days or more are not: **72.3% of them begin between September and January and 17.9% between March and July** ($\chi^2 = 86.1$, $p = 9.5 \times 10^{-14}$; Figure 5a). The gaps long enough to destroy a summer peak are seasonal shutdowns, and they start after the warm season has been recorded, so little is at risk and little

Table 6. Decomposition by outage arm.

panel	arm	cases	base rate	length rule	screen	contrast (95% CI)
development	empirical	14995	7.4%	2.52%	0.38%	−2.13 (−3.45, −1.26)
development	stratified	25709	19.3%	14.58%	6.17%	−8.39 (−9.25, −7.47)
reserved	empirical	4041	4.2%	0.74%	0.11%	−0.71 (−2.36, +0.05)
reserved	stratified	6779	20.8%	16.59%	7.78%	−8.68 (−10.46, −6.51)

aBase rate is the consequential-error rate with no screening; the two rule columns are at 70% retention. The declared five-point effect is met on the pooled and stratified populations and not on the empirical arm.

can be screened out: 4.2% of reserved empirical cases exceed the tolerance with no screening at all.

A counterfactual that holds gap lengths fixed and moves only start month is reported in Figure 5b as an exploratory amendment after correcting year-end truncation in the placement code (`results/wrr_submission_ready/e1_maintenance_recompute/`). Those percentage-point swings are not sealed confirmation and are not used as agency recommendations.

4.2.4 The gain also depends on the agency and on how strictly its values are checked

Pooled numbers conceal two further dependencies. By agency, the sealed transport panel gives −4.3 points on 16 Swedish networks ($[-6.3, -3.2]$), −4.9 on 14 French networks ($[-8.5, -1.5]$) and −6.4 on the two English networks ($[-8.0, +1.3]$, an interval that a two-network sample cannot narrow); the Norwegian secondary stratum gives −6.1 ($[-8.7, -4.6]$). By assurance, the globally checked stratum gives −6.2 points ($[-8.4, -4.5]$) against −4.6 ($[-6.2, -3.3]$) for providers that publish their values unqualified, so part of the pooled gain is carried by records whose quality flags are stronger. The same caution applies to the measurement contract itself: five providers document the daily mean that the parser reads, two publish sub-daily values that the project averages, and four do not document a daily mean at all (`measurement_provenance_dictionary.csv`). Restricting the development panel to the documented-daily-mean stratum gives a contrast of −5.7 points ($[-6.5, -4.9]$, 84

networks) rather than −6.4, and the reserved panel −6.6 ([−7.9, −5.3], 29 networks) rather than −7.1 (`amendments/provider_contract_sensitivity/`). The direction and the significance of the effect are unchanged in every stratum; its size is not a single number.

4.3 The mechanism: hidden support and donor information control the error

Two measurable quantities organise the pattern: the share of the estimand's own support that a gap hides, which enters the error through the influence function of the statistic, and the synchronous anomaly information the surrounding stations carry. Both are visible at decision time, and neither is a restatement of gap length.

4.3.1 A support must be located before it can be conditioned on, so the useful window is wider than the statistic

Two predictions declared for this round were not met, and both failures have one cause. We predicted that the axis restricted to the estimand's own support would be at least as good as any hand-chosen axis at every k, and that its advantage would decay monotonically. Neither holds: the advantage is a hump peaking at a 60-day support, and for supports of 15 days or less the fixed 30-day peak window beats the estimand's own support, by 0.064 AUC at $k = 1$.

Table 7 gives the reason. The gain from conditioning on a window wider than the statistic tracks the year-to-year wander of the realised window's location almost exactly: it is +0.102 AUC at $k = 1$, where the climatological window is the worst estimate of where this year's window will fall and the wander is 16.7 days, and vanishes at $k \geq 90$, where the wander has fallen to 7.4 days and the support is wide enough to contain its own location error. A screening rule therefore does not condition on the support; it conditions on the support *dilated by the uncertainty in the support's location*, and a dilation of roughly two location standard deviations on each side is what the data choose.

One consequence is practical. AUC at the best conditioning window is between 0.85 and 0.86 for every estimand from the annual maximum to the half-year mean, so an agency needs one climatological window of 60 to 90 days rather than one

Table 7. Which climatological window a screen should condition on, development cases.

estimand k (days)	best window (days)	AUC there	AUC on own support	gain	support wander (days)
1	90	0.862	0.760	+0.102	16.7
3	60	0.858	0.766	+0.092	16.3
7	90	0.855	0.778	+0.077	16.5
15	60	0.853	0.802	+0.051	15.3
30	90	0.852	0.827	+0.025	13.7
60	90	0.860	0.855	+0.005	9.4
90	90	0.856	0.856	0	7.4
182	182	0.851	0.851	0	6.9
365	365	0.844	0.844	0	—

aThe diagonal is the estimand's own support; the last column is the median year-to-year circular standard deviation of the realised window's centre.

per metric. A second is uncomfortable: the screen deployed below conditions on a 30-day window and is therefore leaving 0.025 AUC on the table for its own endpoint.

4.3.2 Error grows along hidden support rather than along length, and donors supply anomaly information

The mechanism behind that scale is measurable rather than only plausible. Binning development cases on both properties simultaneously and measuring the growth of error along each with the other averaged out, the hottest 30-day mean grows 3.5× along the missing-days axis and 9.6× along the peak-overlap axis, while the annual mean grows 10.2× and 1.5× respectively: the two ratios swap by roughly an order of magnitude in each direction. For the peak, once the support is in the model, gap length adds nothing measurable — its permutation importance is −0.004 against 0.194 for peak overlap (Figure 6c) — and nested feature sets grouped by network rise from AUC 0.742 with length alone to 0.845 adding the support, 0.858 adding the observed-peak margin and 0.884 adding donor quality (Figure 6b).

Two qualifications belong with that. Length is not irrelevant: holding peak overlap at zero, the consequential-error rate still rises from 0.6% at 1 to 30 days to

8.7% at 121 to 200 days. And in a linear score that already contains warm-season coverage the length term takes a small negative weight, which is collinearity — the two correlate at −0.86 — and not evidence that longer gaps are safer. That correlation is also why a timing-versus-length labelling is wrong: warm-season coverage is a count of missing days restricted to a window, which is a length measure with a domain restriction, and it is the domain restriction and not the timing that does the work.

Donor information enters the residual through quality rather than quantity. Halving the donor set by training-period correlation costs +0.084 °C (95% interval 0.064 to 0.110) when the comparison is re-run with the donor count matched exactly between the two arms, against +0.082 (0.061 to 0.112) under the original construction in which the paired cases can differ by up to seven donors (34% of cases). Replacing donor anomalies with phase-shuffled surrogates while preserving each donor's seasonal cycle costs +0.234 °C (0.149 to 0.299), so donors supply synchronous anomaly information and not merely a seasonal shape.

4.3.3 The calibrated bound is honest enough to interpret

Everything that follows uses a predicted error bound to rank station-years, so the bound is checked before it is used. On the reserved networks the nominal 90% interval covers 90.9% of cases marginally, inside the pre-declared band of 0.87 to 0.93, and 86.3% to 94.9% within every pre-declared stratum (Figure 7a). Median widths adapt with the quantity they are meant to track, from 0.06 °C when the peak window is untouched to 1.87 °C when most of it is hidden.

Those strata pool across networks, and conformal validity assumes an exchangeability that fails between river networks. Coverage inside individual reserved networks has mean 0.904 and median 0.913 but a minimum of 0.778, with five of 36 networks below 0.85. The pooled interval is an honest statement about a population of networks and is *not* a guarantee at a single new site; calibrating on the per-network conformal quantiles at the 0.9 level puts no network below 0.85 for a 1.5-fold widening. The screening results below are identical under every calibration to four decimal places, because the rank order the screen uses is a monotone

transform of the bound, so a reader who distrusts the intervals need not distrust the frontier.

4.4 What it changes: retained error, archive composition, and the decision

A screening rule is judged by the archive it leaves behind, so we close on three consequences: the error among retained station-years at equal retention, which stations survive, and what the two admission rules do to the 2422 records in this archive that really have gaps and have never been decided.

4.4.1 Conditioning on the support cuts the error rate at equal retention

At 70% retention on development cases, the share of retained station-years with a peak error above 0.5 °C is 2.9% for the calibrated screen, 5.6% for a rule on peak overlap alone, 9.2% for the gap-length rule and 15.0% for random retention (Figure 8a). On the 36 reserved networks scored as a retrospective network holdout (mixed prior access; see the prior-access ledger), the screen gives 2.5% against 9.7% for the length rule (Figure 8b): a contrast of −7.13 points, 95% network-clustered interval −8.97 to −5.36, against the five-point effect declared for that freeze. Both declared conditions are met on that holdout, in every donor regime (−6.6, −6.7, −7.2, all excluding zero). An independent sealed USGS confirmation recovers a similar pooled contrast (Section 2.2). The holdout result is not carried by the largest agencies alone: case-equal −7.01, provider-equal −6.29, network-equal −5.82, station-equal −5.54 points, with a median across individual reserved networks of −5.0, 83% of networks favouring the screen, and a worst network of +2.8.

Placed at the operating points the published guidance implies rather than at an arbitrary retention: at the seven-week unfilled limit (49 days, 49.2% retention on the reserved panel) the length rule leaves 5.2% and the screen 0.3%; at the twenty-week imputed limit (140 days, 85.1% retention) the length rule leaves 11.3% and the screen 6.9%. The second of those is more permissive than an earlier version of this analysis assumed — it admits 85% of cases rather than 74% — and the screen's advantage there is 4.4 points rather than 6.2.

Table 8. Reserved panel at 70% retention, across tolerances.

screen	$> 0.25\,^{\circ}$C	$> 0.5\,^{\circ}$C	$> 1.0\,^{\circ}$C	under by $> 1\,^{\circ}$C
outage-length rule	17.2%	9.7%	3.4%	2.0%
peak-overlap rule	9.2%	4.2%	1.2%	0.7%
calibrated screen	7.1%	2.5%	0.4%	0.3%
random	23.9%	14.4%	5.6%	3.7%

aThe last column is the share of retained station-years whose annual peak is underestimated by more than $1\,^{\circ}$C.

Most of the benefit does not require the calibrated machinery (Table 9). Fifty-nine per cent is available from a logistic regression on three variables and 57% from peak-window overlap with no fitting at all; method selection contributes 0.2 points. Fitted on development networks and applied unchanged to the reserved panel, the deployable rule is

$$\text{risk} \;=\; \text{logistic}\big(-1.302 + 2.070\,x_1 - 2.412\,x_2 - 0.00187\,x_3\big), \tag{4}$$

where x_1 is the share of the station's climatological hottest-30 window the gap hides, x_2 the share of May-to-September days that remain observed, and x_3 the length in days of the longest consecutive gap. It gives 4.2% against the length rule's 9.6% at 70% retention, a contrast of -5.44 points (-6.85, -3.96), and cuts severe underestimation of the peak from 1.9% to 0.6%.

4.4.2 The incumbent is not handicapped, including by the absence of the field's own reconstruction

A screening comparison can be rigged by pairing the incumbent with a weak reconstruction, so the incumbent is paired here with every reconstruction in the study, including one nobody can deploy (Table 10).

The screen beats the incumbent armed with a reconstruction nobody can run, and the reconstruction paired with the incumbent throughout is the strongest of the deployable options rather than the weakest. Air-driven reconstruction is reported twice, because the earlier label was wrong: air-temperature regression remains a workhorse for data-sparse basins (Sanikhani et al., 2026; You et al., 2026), and the

Table 9. Component ablation, development cases at 70% retention, each variant re-run through the full out-of-fold pipeline.

screen	error rate	points vs the length rule
outage-length rule (incumbent)	9.24%	—
three-variable logistic regression	5.53%	−3.71
peak-overlap alone, no fitting at all	5.62%	−3.62
three-feature gradient boosting	4.89%	−4.35
geometry and timing only (8 features)	4.86%	−4.38
drop donor features	3.78%	−5.46
drop the observed-peak margin	3.13%	−6.11
full screen	2.93%	−6.31
full screen, screening only (no method choice)	3.11%	−6.13

library's air ridge with a fixed penalty ($\alpha = 1.0$, seven lags, whole-year fit, penalised) was described as "in the manner of Beaufort et al. (2022)" but shares only its input with that paper, and is now labelled as what it is. Reimplementing the cited method faithfully — per-station OLS of summer water temperature on a right-aligned moving average of air temperature, 2–10 day windows chosen per station by R^2, fitted on July–August — gives 12.76% against the library ridge's 15.16% (paired difference −2.40 points, $[-3.20, -1.62]$, 1200 network-clustered draws), and both remain worse than per-case selection at 9.00% (`amendments/faithful_baseline/`). The axis conclusion does not depend on the estimator library: the timing-minus-length AUC contrast is +0.070 with the old forcing and +0.081 with the new. The screen uses the new information, selecting the air-driven estimator for 20.0% of cases against 0.07% before, and donor regression for 53.1%.

One result here was not expected: the *fixed* information-conditioned policy gets slightly worse at donor-less stations when air becomes available, from 0.340 to 0.364 °C.

Table 10. Development cases at 70% retention, with air forcing available throughout.

screen	reconstruction	error rate above 0.5 °C
length rule	information-conditioned policy (used throughout)	9.00%
length rule	always library air ridge, $\alpha = 1.0$, else climatology	15.16%
length rule	always faithful Beaufort et al. (2022) climate correction	12.76%
length rule	always donor ridge, else climatology	9.57%
length rule	always smooth climatology	19.79%
length rule	oracle over methods, not deployable	3.62%
calibrated screen	deployable estimators it selects itself	2.87%

4.4.3 On records that really have gaps, the two rules disagree on one year in six

Every number above is computed on a complete station-year with a synthetic hole punched in it, which is the only way to have a truth. The same archive contains 2422 genuinely incomplete station-years, a third of the total, which carry no truth and have never been decided. Three things can be asked of them without inventing one (Figure 9).

They are not a marginal population. The median incomplete year is missing 9 days and the 90th percentile 128; 28.7% of them have at least one day of the station's climatological peak window missing, and 261 have more than half of it missing.

The two rules disagree on a sixth of them. At the seven-week unfilled limit the length rule admits 83.2% and a peak-window rule at 10% hidden admits 81.2% — near-equal retention, which is the comparison this paper insists on — and they **disagree on 15.1%** of station-years: 207 are admitted by the incumbent and rejected on the support, and 158 are rejected by the incumbent and admitted on it. At the twenty-week imputed limit the incumbent admits 91.7% and the disagreement is 11.6%, of which 267 are in the dangerous direction and 13 in the wasteful

Table 11. Archive composition under each admission rule, on the 2422 real incomplete station-years.

rule	admitted	any day hidden	half hidden	wholly hidden
repair everything, no screening	2422	696	261	186
length $\leq$ 140 d, with imputation	2221	507	82	17
length $\leq$ 49 d, unfilled	2016	437	45	2
peak window $\leq$ 10% hidden	1967	241	0	0

aBoth published limits are those of Johnson et al. (2021). The last three columns need no truth: they count admitted station-years in which any day, more than half, or all of the climatological peak window is unobserved. A wholly hidden climatological window is a proxy for a wholly unobserved realised peak, with a measured positive predictive value of 0.74 on sealed designed cases (Section 4.4.6).

one. The disagreement is present in every provider with more than 20 incomplete station-years, from 11.1% to 25.7%.

What the disagreement costs can be counted without a truth (Table 11). The published guideline agencies are most likely to apply, the twenty-week limit that exists because imputation is assumed, admits 17 station-years whose *climatological* peak window was never observed at all and 82 whose peak window is mostly gone. That count is a proxy and its error rate is measurable: on the sealed designed cases, where the realised peak is known, a wholly hidden climatological window carries a positive predictive value of 0.74 (network-clustered interval 0.71 to 0.76) for a wholly hidden *realised* peak, with sensitivity 0.87 and only 3.1% of proxy-clear cases hiding the realised peak. Transported to the 17 flagged station-years, that implies about 12 to 13 whose realised annual peak was wholly unobserved and about 4 whose proxy fired while the realised peak was visible. A risk model fitted on development cases from decision-time features only, and checked for calibration on the reserved panel first (largest decile miscalibration 6.2 points, and under 1.6 points in nine of ten deciles), predicts a consequential-error rate of 0.98% among the years the unfilled limit admits, 1.85% among those the imputed limit admits, and 0.26% among those the peak-window rule admits at essentially the same retention. That last figure is an extrapolation from scored cases to unscored records and is labelled as one; Section 4.4.6 tests the exchangeability it assumes.

At the level of a regional aggregate the choice of rule barely matters: admitting incomplete years at all raises the archive's mean annual peak by about half a degree, and the four rules differ by less than 0.1 °C (+0.52, +0.51, +0.53 and +0.45) and by 0.4 points in the share above 20 °C (+4.55 to +4.17).

The same is true of a multi-year slope. Theil–Sen slopes of the annual peak have RMSE 0.143 $°C\,yr^{-1}$ on recorded days, 0.044 with reconstruction and no screening, 0.048 under length screening and 0.041 under the screen: the dominant step is again whether the record is repaired. Eleven complete years per station is in any case too short to resolve a decadal trend, so we report the slope error as a diagnostic of the reconstruction and make no trend claim.

Per station-year the choice does matter, and Section 5.3 follows it into the judgement an agency actually makes.

4.4.4 Conditioning on difficulty removes mountain catchments unless the threshold is applied within strata

A rule that lowers the average error by discarding hard station-years has bought accuracy with representativeness, and in this application the hard sites are not a random subset: Section 4.4.5 showed they are high-elevation, steep, donor-poor catchments, which are the cold-water headwaters the summer endpoint exists to describe.

At 70% retention the calibrated screen's per-station retention correlates with site elevation at Spearman $\rho = -0.26$ ($p = 0.011$) on development and -0.27 ($p = 0.077$) on the reserved panel, while the incumbent length rule is elevation-neutral at $\rho = -0.05$ ($p = 0.60$) and -0.10 ($p = 0.52$). On the reserved panel the screen retains 77.7% of station-years in the lowest elevation tertile and 66.9% in the highest, a spread of 10.8 points against 1.7 for the length rule (Figure 10b). By donor regime the same pattern appears on development cases: 57.1% at donor-less stations against 72.5% at donor-rich ones, a 15.4-point spread, while the length rule tilts the other way.

The distortion is cheap to remove (Table 12). Applying the screen's threshold within stratum rather than globally costs 0.31 points of consequential-error rate

Table 12. Error against representativeness, reserved panel at 70% retention.

rule	error rate	ρ with elevation	donor spread	retention SD
outage-length rule	8.81%	−0.10	4.5 pts	0.075
peak-overlap rule	3.75%	+0.16	2.4 pts	0.104
calibrated screen, global threshold	2.53%	−0.27	4.3 pts	0.149
screen, balanced within donor regime	2.52%	−0.27	0.1 pts	0.148
screen, balanced within elevation tertile	2.85%	+0.08	4.6 pts	0.148
screen, balanced within both	2.84%	+0.07	0.1 pts	0.144

aRetention SD is the standard deviation of per-station retention across stations; a larger value means the rule treats stations more unevenly.

on development cases and 0.30 on the reserved panel and flattens both distortions. Stratifying on donor regime alone does not fix the elevation tilt and stratifying on elevation alone does not fix the donor tilt, so an agency that cares about both must declare both.

4.4.5 Where it is hard: catchment attributes explain a modest share of the error

The account so far is a property of estimators and estimands and would hold for any time series. For the 140 United States stations matched to GAGES-II across 57 networks, we regressed donor synchrony and completed-year error on basin attributes with the expected direction of each effect declared before the regressions were run (Supporting Information Figure S7). Site elevation and basin slope both reduce donor synchrony (Spearman $\rho = -0.29$, 95% CI −0.46 to −0.08, and $\rho = -0.31$, −0.51 to −0.07), and site elevation raises both completed-year error ($\rho = +0.24$, 0.06 to 0.43) and the consequential-error rate ($\rho = +0.25$, 0.08 to 0.43).

The groundwater prediction taken from the literature is not reproduced, and the reason is instructive. Johnson et al. (2021) report that filling can harm groundwater-influenced sites, which predicts larger completed-year error at a high

baseflow index. Baseflow index does reduce donor synchrony in the predicted direction, from 0.70 in the lowest tertile to 0.59 in the highest, although the rank correlation interval includes zero ($\rho = -0.19$, -0.42 to 0.10). But the error moves the other way: splitting at the median baseflow index, high-baseflow stations are $-0.043\,°C$ easier than low-baseflow stations (95% CI -0.081 to -0.004). Across baseflow tertiles the mean annual peak falls from 22.7 °C to 18.1 °C ($\rho = -0.47$, -0.60 to -0.26) and its interannual standard deviation from 0.98 to 0.76 °C (Supporting Information Figure S7c), so two effects act in opposite directions and the second is larger.

Regulation does not make station-years harder for this endpoint: reference and non-reference basins differ by $+0.004\,°C$ (95% CI -0.051 to $+0.093$). Catchment attributes explain a modest share of the variation, a standardised linear model reaching $R^2 = 0.22$ for donor synchrony and 0.13 for completed-year error, and no attribute predicts how much a screen gains: every correlation between a basin attribute and the per-station screening benefit has an interval containing zero. Whether an attribute predicts what a screen *keeps* is a different question, and Section 4.4.4 answers it differently.

4.4.6 The conclusion survives instrument error, a stricter fitting protocol, and transfer across agencies

Missingness is weakly informative, and the effect is bounded well below the tolerance. Every case transplants a mask onto a year whose values were observed, which is valid only if the mask is ignorable in the sense of Rubin (1976); loggers are lost to floods, buried by ice and fouled in low flow, and each of those coincides with anomalous conditions. Across the 6472 real gaps, the mean anomaly of the seven days preceding a gap, relative to the station's own day-of-year climatology, is $+0.018\,°C$ for gaps of a week or less ($p = 0.44$), +0.038 for 8 to 29 days ($p = 0.56$), -0.002 for 30 to 89 days ($p = 0.98$) and -0.215 for 90 days and longer ($p = 0.081$). Donors recording through a gap run $+0.063\,°C$ warm during short gaps, detectable at this sample size ($p = 0.017$) and 3% of the within-station anomaly scale. The design is therefore defended on the magnitude of the effect rather than on its absence, and the class that warrants a limitation is gaps of three months and longer. A pattern-mixture check turns that defence into a bound: shifting every filled day by

δ °C moves the completed year's peak by δ times the share of the realised peak window the gap hid, and across $\delta \in \{-1, -0.5, -0.25, 0, +0.25, +0.5, +1\}$ °C the screen keeps its advantage at equal retention with a network-clustered interval excluding zero in all three populations (development, the retrospective reserved panel, and the sealed USGS panel); no bias inside that grid erases it (`amendments/mnar_pattern_mixture/`). The assumption is bounded rather than verified, which is the strongest statement these records support.

Three further checks move the numbers and not the conclusion. Instrument error in these archives is systematic and time-varying as well as random (Jackson et al., 2025), which is why the check below is repeated under correlated, drifting and step-like error and not only under white noise. Perturbing the reference series by 0.15 °C of normal noise moves the development contrast from 5.98 to 6.03 points, the reserved contrast from 6.21 to 6.22 points, and the mean absolute error from 0.2361 to 0.2363 °C; the paired change is +0.049 points (−0.036, +0.150) on development and +0.013 (−0.175, +0.185) on the reserved panel, so the check no longer reports a reduction in the advantage (an earlier version evaluated the two arms with two different reconstruction choices). Correlated error displaces the 30-day endpoint harder at the same daily scale and still leaves the contrast within its interval: an AR(1) process with $\rho = 0.7$ by 2.25 times and a mid-year 0.3 °C step by 2.12 times, giving 5.95 and 6.21 points (development) and 6.19 and 6.43 (reserved). A constant annual bias of ±0.5 °C is an exact endpoint shift, so it moves the contrast by at most 1.0 point (largest change interval 0.35 to 1.82 points, on the reserved panel at −0.5 °C) while raising the mean absolute error to 0.30 °C. Restricting estimators to years before the gap raises reconstruction error from 0.242 to 0.270 °C on a panel that falls to 33633 development cases, and moves the contrast from −6.35 (−7.12, −5.52) to −5.92 points (−6.72, −4.99) with coverage 0.906. Under leave-one-provider-out the contrast is negative with an interval excluding zero for all nine agencies, from −11.6 points (Finland) to −1.2 (Switzerland), with conformal coverage between 0.82 and 0.96.

5 Discussion

Using 5114 complete river station-years, 51524 designed outages and two sealed panels, we have shown that the error of an annual thermal extreme is governed by

where a gap falls rather than by how long it is, that an annual mean reverses that ordering, and that screening on the hidden share of the statistic's support leaves a materially better archive than a completeness threshold at equal retention. The result agrees with the standing guidance on the estimands that guidance was derived for, and disagrees with its transfer to extremes; the disagreement is one of conditioning axis, not of arithmetic. Two mechanisms account for it, one of which the data can measure and one of which they cannot.

5.1 Two mechanisms explain the reversal, and one of them is not in the data

The transition measured in Table 4 is not a property of rivers, of the estimators, or of the screening rule. It is a property of the functional being computed, and it follows from where that functional puts its weight. A statistic whose influence is concentrated on a few days is destroyed by losing those days and is almost unaffected by losing any others; a statistic whose influence is spread evenly over the year tends to degrade with how many days go missing and is less sensitive to which. Between those two limits the influence is neither concentrated nor uniform, and the discriminating power of a narrow conditioning window falls as the weight it captures falls. On this panel's exploratory width sweep, a fixed 30-day window stops beating a whole-year count near a three-month support; that numerical crossing is panel- and climate-dependent, not a universal constant.

Two measurements distinguish that account from a re-description of the data. First, the gradient ratios swap by roughly an order of magnitude in each direction between the peak and the annual mean, which is a direct measurement of how error grows along each axis rather than an inference from discrimination. Second, and more decisively, the account makes a prediction about a case it was not built on. The sine-fit phase has an influence function with two broad lobes a quarter-year from the peak, so the account predicts that peak-window overlap should be a poor axis for it — and it is, at 0.583 against 0.836 for gap length. A description fitted to the extreme-versus-average contrast would not have predicted that.

The same account explains where it failed. We predicted that conditioning on the estimand's own support would always be best, and it is not, because the support

has to be estimated and the estimate wanders. Once the wander is measured the failure becomes a quantitative statement rather than an exception: the gain from widening the conditioning window tracks the wander from 16.7 days at $k = 1$ to 7.4 days at $k = 90$ and disappears when the support becomes wide enough to contain it. This is the part of the result we would expect to transfer to other variables and other climates, because nothing in it is specific to water temperature: any estimator conditioned on an estimated support inherits the uncertainty of that support.

5.2 Where this agrees with the standing guidance, and where the disagreement lies

An earlier framing of this work held that the incumbent rule gets an ordering backwards. Section 4.2.2 shows that reading is wrong on the guidance's own terms. Johnson et al. (2021) derive their limits for sine-fit mean, phase and amplitude, and for those three estimands a completeness threshold is the correct axis or within 0.02 AUC of it. What we correct is not the threshold but the habit of carrying it across to an estimand it was not derived for.

That reading also resolves the anomaly the guidance itself reported. Its authors observed that the timing of consecutive missing data had "inconsistent effects ... among regions, years, and individual parameters" on annual thermal-signal estimates. Their three estimands have three different influence functions — a constant, two lobes at the extremes, two lobes at the transitions — and only one of the three has weight concentrated narrowly enough for a timing effect to appear. Observing all three at once, inconsistency is the only thing one could see. We do not contradict the observation; we decompose it, and we can now say in advance which of their parameters should show a timing effect and which should not.

A smaller result points the same way and was not expected. The fixed information-conditioned policy gets slightly *worse* at donor-less stations once air forcing becomes available, because its preference order was written when air was rare and it now prefers an air regression to a climatology at sites where the climatology is better. A fixed ladder over estimators is not robust to a change in what is available; a per-case selection is. That is an argument for the method of this paper rather than against it, and we report it because it was the opposite of what we expected.

The disagreement with the gap-filling literature is of a different kind and is not resolved by our result. Beaufort et al. (2022) fill every record alike, and on our panel pairing the incumbent rule with that treatment makes it worse rather than better (Table 10). That is not evidence against air-temperature regression, which is the best available estimator at many of our stations and which the screen selects for 20.0% of cases. It is evidence that a reconstruction chosen once for a whole archive and a reconstruction chosen per station-year are different objects, and that the quantity a screening comparison must hold fixed is the second.

5.3 What changes in the judgement an agency actually makes

The archive-level aggregate is the wrong place to look for the value of a screening rule, because it averages over the station-years where the call is easy. A criterion-exceedance judgement is the right place, because it is made one station-year at a time and it is where a wrong peak becomes a wrong decision. On reserved networks the misclassification rate of a 20 °C exceedance is 6.86% when the statistic is computed on recorded days, 1.97% when every year is reconstructed and not screened, 1.53% under length screening and 0.36% under the screen. Restricted to the years where the call is close the four figures are 16.84%, 9.50%, 7.26% and 1.92% within 1.0 °C of the threshold and 19.10%, 13.12%, 10.24% and 3.38% within 0.5 °C, so the advantage is proportionally larger exactly where it is needed. The bias in the count of years above 20 °C falls from −0.76 years per station under recorded days and −0.32 under length screening to +0.06.

That result is not an artefact of the threshold chosen. Swept from 14 to 24 °C the screen's misclassification stays between 0.18% and 0.37% while the length rule stays between 0.84% and 1.53%. The sweep matters because no single number is the criterion: regulatory practice uses a seven-day average of daily maxima with criteria differing by several degrees between jurisdictions and designated uses, 16 °C for core cold-water salmonid habitat and 13 °C for salmonid spawning (Washington State Department of Ecology, 2024), which is why the seven-day endpoint is reported throughout and why the threshold is swept rather than asserted. The illustration is not a compliance test, and we can now say by how much it misses one: on the 1591 station-years of the sealed USGS panel that publish both statistics, the regulatory seven-day average of daily maxima sits a median +1.57 °C above our seven-day

average of daily means (5th to 95th percentile +0.49 to +4.43), and at criteria of 16, 18 and 20 °C the two endpoints disagree on 5.5%, 4.3% and 10.4% of station-years — every disagreement a missed exceedance and none a false one (`amendments/regulatory_daily_max/`). A threshold shift matches counts but not decisions, because the offset is nearly constant (slope −0.002 per °C). An agency applying a daily-maximum criterion must compute it from daily maxima, which the archive publishes for 90 of the 91 sealed stations.

For a judgement of this kind, screening is not the best use of the calibrated object. Discarding a station-year is a blunt response to uncertainty about a single call: screening at the 0.5 °C tolerance judges 67.5% of years at 0.36% misclassification, whereas abstaining only where the interval straddles the threshold judges 92.7% at 0.42%, and doing both judges 65.7% at 0.08%. Keeping 93% of station-years rather than 68% at essentially the same error rate matters to anyone assembling a regional archive. The two actions answer different questions — screening decides what enters an archive, abstention decides what one judgement may say — and the same calibrated object serves both, which is an argument for calibrating rather than for screening.

5.4 What an agency should change, and in what order

The first recommendation is to treat screening as estimand-specific. An agency assembling warm-season extremes from maximum gap length should trial conditioning on how much of a station's climatological warm window the gap hides, and should keep length-based rules for annual means and the sine-fit regime parameters for which standing guidance was derived. Cold extremes are not an exception to this: the support axis has to be defined on the cold season, and when it is, it beats the length rule for the coldest 30-day mean on the development panel; a warm-season support axis is the wrong tool there, which is what the reversal in Table 3 shows. On the retrospective reserved holdout, substituting a support-aware screen takes the consequential-error rate from 9.7% to 2.5% at equal retention on pooled designed outages; the empirical-arm gain is not resolved, so this is not yet a claim about the networks' natural failure calendar.

Maintenance-calendar pricing that previously appeared to show autumn servicing gains of several percentage points is withdrawn from recommendations until the cross-year outage placement is recomputed: the earlier counterfactual truncated late-year gaps at day 366 while still labelling them by their nominal length. Until that recompute is confirmed on networks outside the development programme, schedule changes remain a hypothesis suggested by autumn-heavy long outages, not a verified error-rate benefit. It cannot be replaced by an observed counterfactual on this archive, and we checked rather than assumed that: for the 200 USGS stations of the incomplete-year population we classified all 2921 real interior gaps and 68180 missing days against the provider's own daily-value qualifiers, and no cause-naming code — maintenance, equipment, ice, seasonal or discontinued — occurs once. What the qualifiers do support is an archival decomposition: 94.9% of missing days are days the provider published no record for at all (interval 88.7 to 98.2), 4.8% were published only under a different statistic of the same parameter, and 0.3% were published for the target statistic and dropped by the archive (`amendments/outage_causes/`). A rule that excludes planned-maintenance gaps and keeps equipment failures is therefore a no-op on 0 of 1361 station-years, while re-ingestion would recover five station-years (all 2024–2025) whose target values the live service now serves. Redundancy investments similarly need cost and failure-mode evidence beyond synchrony correlations on the opened panel.

Thresholds should be applied within strata when difficulty correlates with elevation or donor poverty, or the archive can quietly lose mountain stations. Nothing here applies to ungauged sites, because locating support uses a station's own history. A pooled conformal interval is a statement about a population of networks, not an automatic guarantee at a single new site; cluster-aware calibration must be chosen on calibration data only and validated on networks outside the development programme before it is sold as site-level coverage.

5.5 What generalises, and what does not

Three parts of the result are portable. The governing quantity of Equation (3) is defined wherever a derived statistic is computed from an incomplete record, and its ingredients — an influence function and a mask — are available in any such setting. The protocol that measures the transition is portable in the same way:

compute the statistic over a family of support widths, measure how far the support's location moves between periods, dilate by that amount, and condition on the result. And the finding that a screening rule must be specified together with its strata is a statement about selection, not about temperature.

Three parts are not. The numerical crossover near three months depends on how concentrated a river's thermal year is and on how far its peak wanders between years; both are properties of this climate and this variable. The operational gain is regime-dependent: the sealed USGS confirmation recovers a pooled contrast of -7.9 points but the empirical arm only -1.7, and the multi-agency transport panel, while negative as a whole (-4.6, $[-6.5, -3.3]$), carries an empirical arm of -1.9 ($[-3.3, -1.0]$) — real, but far smaller than the designed-outage figure an agency might quote. Catchment associations rest on 140 of 204 stations in one country and should be read as attribute correlations on that subset, not as a geology-versus-terrain law.

One boundary deserves naming separately, because it limits interpretation rather than transfer. The mechanism argument is supported by an influence-function derivation, by measured error gradients, by a continuum with a located crossover and by an out-of-sample prediction about the phase. It is not supported by an independent observation of the internal quantity itself, because the hidden mass of Equation (3) is by construction unobservable on the records it matters for. A different representation that reproduced the same gradients would not be excluded by our evidence.

5.6 What the remaining uncertainty changes, and what it does not

Uncertainty enters this study in four places, and it is worth separating what each one moves. Measurement error in the reference series changes the numbers in the third decimal and changes neither the mechanism nor the recommendation: $0.15\,^{\circ}\mathrm{C}$ of noise moves the reserved contrast from 6.21 to 6.22 points, and time-correlated error displaces the endpoint twice as hard while still leaving the contrast inside its interval. Fitting protocol changes the numbers by about half a point and changes nothing else: restricting estimators to prior years moves the contrast from -6.35 to -5.92. Neither of these is capable of overturning a conclusion.

The other two are. The choice of population changes the practical conclusion completely: on the arm built from the gap geometry these networks actually experience the effect is -0.71 points with an interval containing zero, and any reader who needs a number for their own archive needs the arm that matches their gap regime rather than the pooled figure. And the absence of an independent observation of the internal quantity limits the mechanism judgement, as the previous section states: we can show that conditioning on an estimate of the support discriminates better, and we cannot show that no other representation would do as well.

Two further limits belong here because they are specific rather than generic. Missingness is weakly informative and the transplant design is defended on the magnitude of that effect rather than on its absence, which makes the defence weakest for gaps of 90 days and longer, where the leading flank runs 0.215 °C cold. And the archive-level consequence is small: the choice of rule moves a regional mean annual peak by less than 0.1 °C. We report that because it deflates a claim we might have made. At the level of a regional aggregate the decision that matters is whether to repair; the rule matters per station-year, and it matters for who is in the archive.

6 Conclusions

Using 5114 complete station-years from 146 river networks, 51524 designed outages, and sealed panels scored once outside the development programme, we quantified what an incomplete year can still support. Three answers follow.

First, the governing property of a gap is its position, not its length: the hottest 30-day mean grows 9.6× along the hidden-support axis and 3.5× along the length axis, the annual mean reverses the two, and a frozen screen conditioned on hidden support cuts consequential errors for the summer peak on the retrospective holdout from 9.7% to 2.5% at 70% retention (-7.1 points, $[-9.0, -5.4]$), with -7.9 ($[-10.2, -5.8]$) on 39 sealed USGS networks and -4.6 ($[-6.5, -3.3]$) on a pre-registered multi-agency panel of 32 networks.

Second, the gain is conditional in three ways that a pooled number hides: it depends on the support width of the estimand, it collapses towards -0.7 to -1.9 points under the gap calendar these networks actually keep, and it is smaller (-4.6 against -6.2 points) where providers publish unchecked values. Completeness thresh-

olds remain the correct axis for annual means and the sine-fit parameters, which is where the standing guidance was derived.

Third, the practical consequence is an estimand-specific admission decision rather than a new universal threshold. An agency assembling warm-season extremes should screen on how much of the climatological peak window is hidden and should apply the threshold within strata, because difficulty-based screening otherwise thins high, steep and donor-poor catchments. Autumn maintenance benefits and "annual peak never observed" counts are not retained as verified effects: the former needs an outage-cause record this archive does not publish, and the latter currently counts wholly masked *climatological* windows, whose positive predictive value for a wholly unobserved realised peak is 0.74.

Open Research Statement

Daily water-temperature records are from nine public providers (ARSO, CHMI, FOEN, GKD Bayern, LUBW, NVE, Rijkswaterstaat, SYKE and the USGS), plus SMHI and the UK Environment Agency for the transport panel; none required a data-use agreement. Nine of the eleven publish under a named open licence (US public domain; CC BY 4.0 for CHMI, GKD Bayern and SYKE; OGL v3.0 for the UK EA; opendata.swiss terms for FOEN; NLOD/CC BY for NVE; CC BY 4.0 SE for SMHI; Licence Ouverte for Hub'Eau). Two state terms without naming a licence (ARSO requires attribution on publication; Rijkswaterstaat requires source and retrieval date). One, LUBW, reserves copyright: its raw values are therefore *not* redistributed with this paper, and the seven LUBW networks are available only through LUBW's own service; the analysis reads them locally and reports derived statistics. Licences, acquisition routes and verbatim evidence are in `metadata/data_rights.csv`, `measurement_provenance_dictionary.csv` and `current_data_acquisition.csv`. Station coordinates are from those providers' own catalogues, the USGS NWIS site service, and ARSO's public station feed. Daily 2 m air temperature is NASA POWER (MERRA-2) (NASA Langley Research Center, 2026). Basin characteristics are GAGES-II (Falcone, 2011).

All analysis code, derived artifacts, figures, the editable draw.io sources of the schematic figures and the frozen pre-registration are in the accompanying reposi-

tory. A claim registry (`claim_registry.csv`) records each headline claim's artifact, endpoint, unit, denominator, evidence type and whether it may enter the abstract or conclusions. An automated audit checks a declared set of archived manuscript numbers against artifacts; it is not yet a full semantic binding of every live TEX numeral. Input-file hashes for the 146 study networks are in `results/wrr_submission_ready/current_data_manifest.csv`. Source data for the figures are deposited with the analysis: the schematic figures are draw.io files (`figures/v17/drawio/`), and every data figure names its source artifact in its caption, with the plotted values and their input hashes also written to `figures/wrr/`. A permanent archive of the repository and a DOI will be deposited on acceptance.

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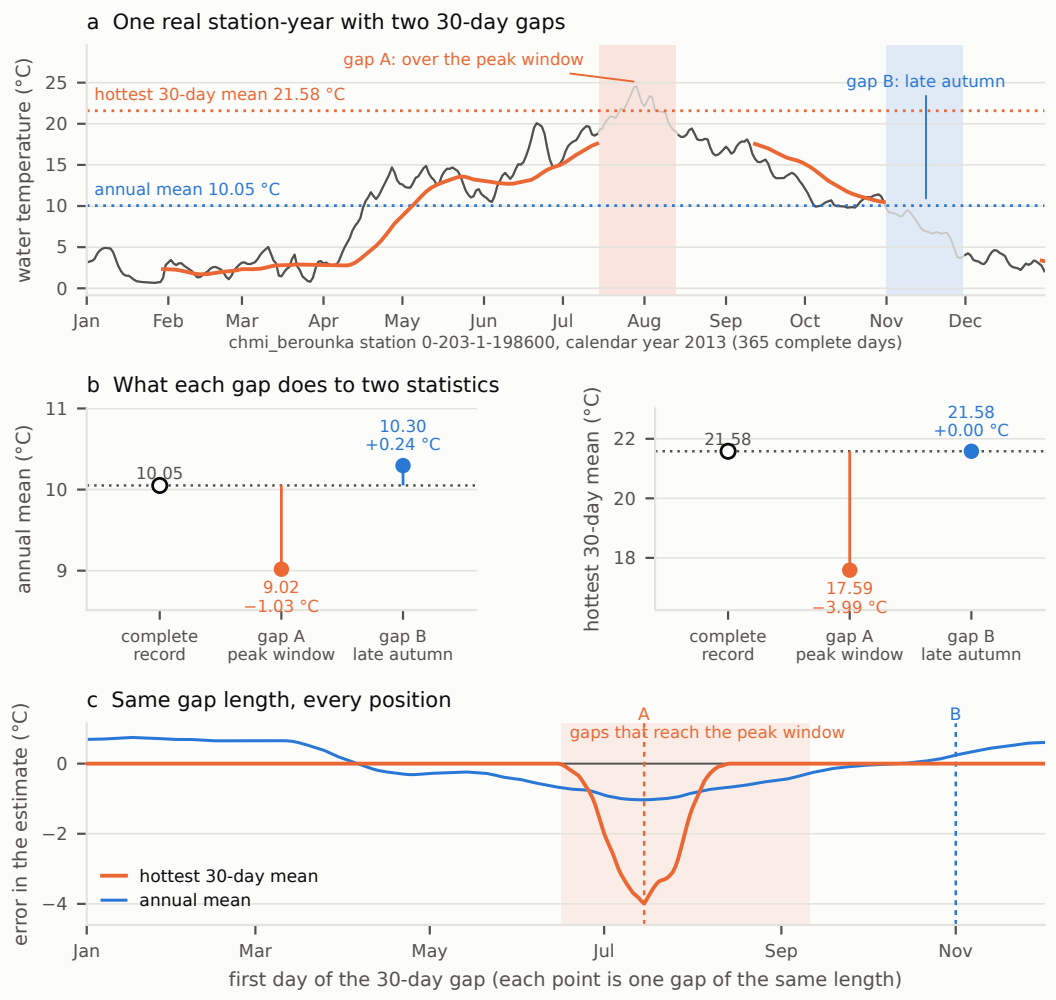


Figure 2. Two statistics, one record, two equal-length gaps. (a) A complete annual water-temperature record (CHMI Berounka, station 0-203-1-198600, 2013; n = 365 days, no missing day) with two designed 30-day gaps: gap A over the window that achieves the complete-record hottest 30-day mean (15 July to 13 August) and gap B in late autumn (1 to 30 November). The curve is dark where the gap leaves it observed and faint where the analyst would not see it. (b) Each gap's effect on each statistic, in °C: dots are estimates from the days the gap leaves, open dots and dotted lines the complete-record truth (annual mean 10.05 °C, hottest 30-day mean 21.58 °C). The panels use different vertical scales and neither starts at zero, so the printed values carry the comparison: gap A moves the hottest 30-day mean by −3.99 °C while the same length in late autumn leaves it exact, whereas the annual mean moves by −1.03 and +0.24 °C. (c) All 336 possible 30-day gap starts of the same year, showing the error each position induces in the hottest 30-day mean (orange) and the annual mean (blue); the shaded band holds starts whose window reaches the peak window. An illustration on one real record, not a result: no model, no population estimate, no test. Source data: `figures/wrr/f1_example_data.json` and `f1_example_sweep.csv`.

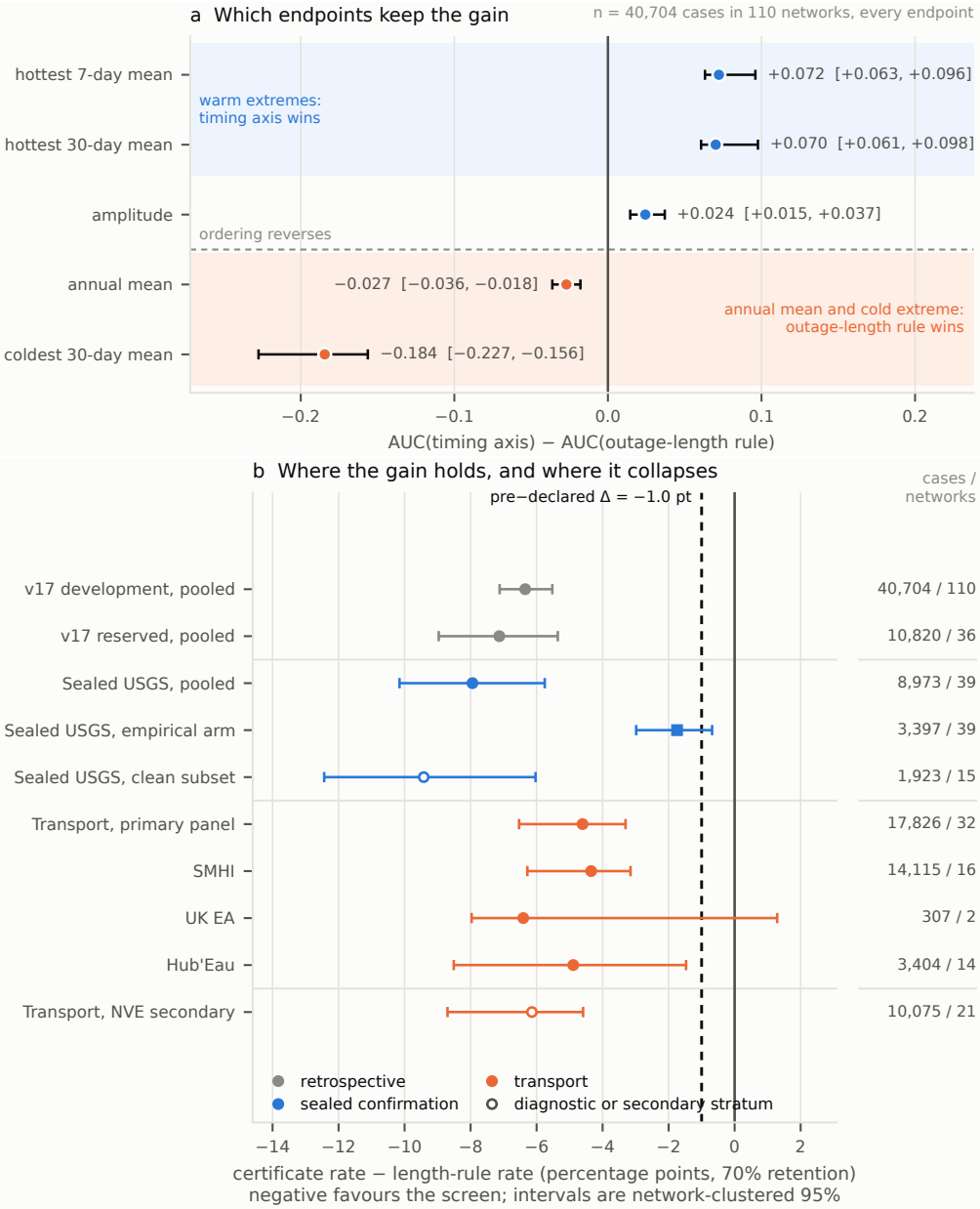


Figure 3. The same screening gain, ordered two ways. (a) Discrimination of a consequential error by endpoint, as the AUC difference between the best timing axis and the outage-length rule, with network-clustered 95% intervals; positive means the timing axis wins. The five endpoints reverse: the two extremes sit on the positive side, the annual mean and cold extreme on the negative side, with the amplitude between them. (b) The pooled contrast (certificate rate minus length-rule rate, percentage points; negative favours the screen) for every evidence panel in this study: retrospective development and reserved panels (grey), the sealed USGS confirmation with its arm and clean-subset variants (blue), and the pre-registered multi-agency transport panel with its providers and the NVE secondary stratum (orange). The dashed line is the pre-declared $\Delta = -1.0$ point threshold; intervals are network-clustered 95%. Source data: `figures/wrr/f2_heterogeneity_data.json`.

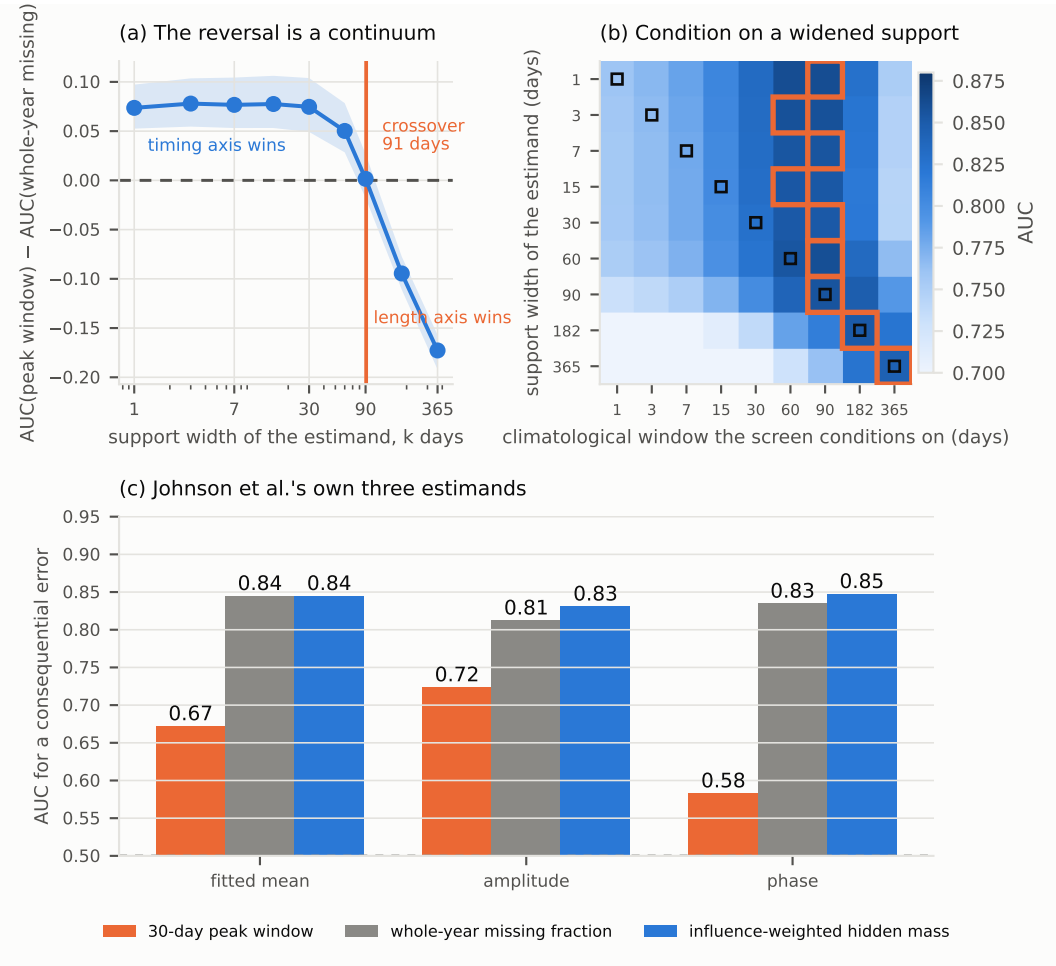


Figure 4. The transition is continuous, and it is the estimand's support that decides. Exploratory results from the opened panel, not the sealed USGS confirmation. (a) Advantage in AUC of a fixed 30-day peak window over whole-year completeness, against the support width k of the estimand (1 to 365 days), development cases; the crossing is an interval, 77 to 107 days, and the sign is reversed for $k = 182$ and $k = 365$ days. (b) AUC of each conditioning axis for each estimand, so that the reversal can be read column by column; intervals are network-clustered 95%. (c) The same axes for the three sine-fit parameters of the standing guidance, each evaluated at 0.47 of its own interannual standard deviation. Because the support of a year must be located before it can be conditioned on, the useful conditioning window is wider than the statistic itself, which is why a fixed narrow window turns over. Source data: `results/v18_closure/e6_support_width/` and `e3_sine_endpoints/`.

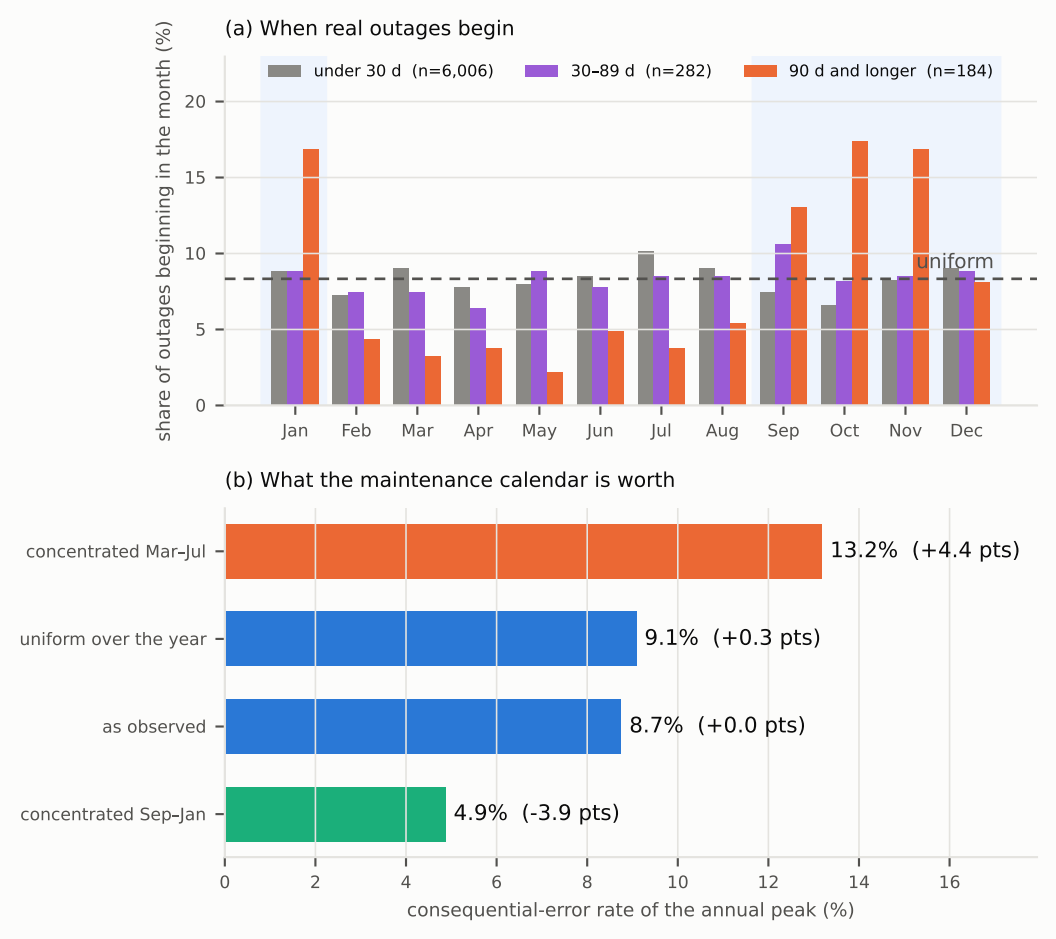


Figure 5. The gap calendar these networks actually keep. (a) Start month of the 6472 real interior gaps found in complete years, split by length class: 5246 gaps of 1–7 days, 760 of 8–29 days, 282 of 30–89 days and 184 of 90 days or longer (median gap 2 days, 95th percentile 43 days). Long outages begin in autumn and winter, which is where the peak window is not; 55.8% of them ($[50.9, 60.9]$%, 99 network clusters) hide less than half of the climatological peak window. (b) Exploratory maintenance-calendar counterfactual: consequential-error rate for 32 240 outage placements drawn from the observed calendar, the maintenance calendar, and a uniform calendar after the multi-year placement correction of the amendment; differences are within the interval, so no maintenance benefit is claimed. Descriptive and exploratory, not a sealed result. Source data: `results/v18_closure/e1_outage_seasonality/`.

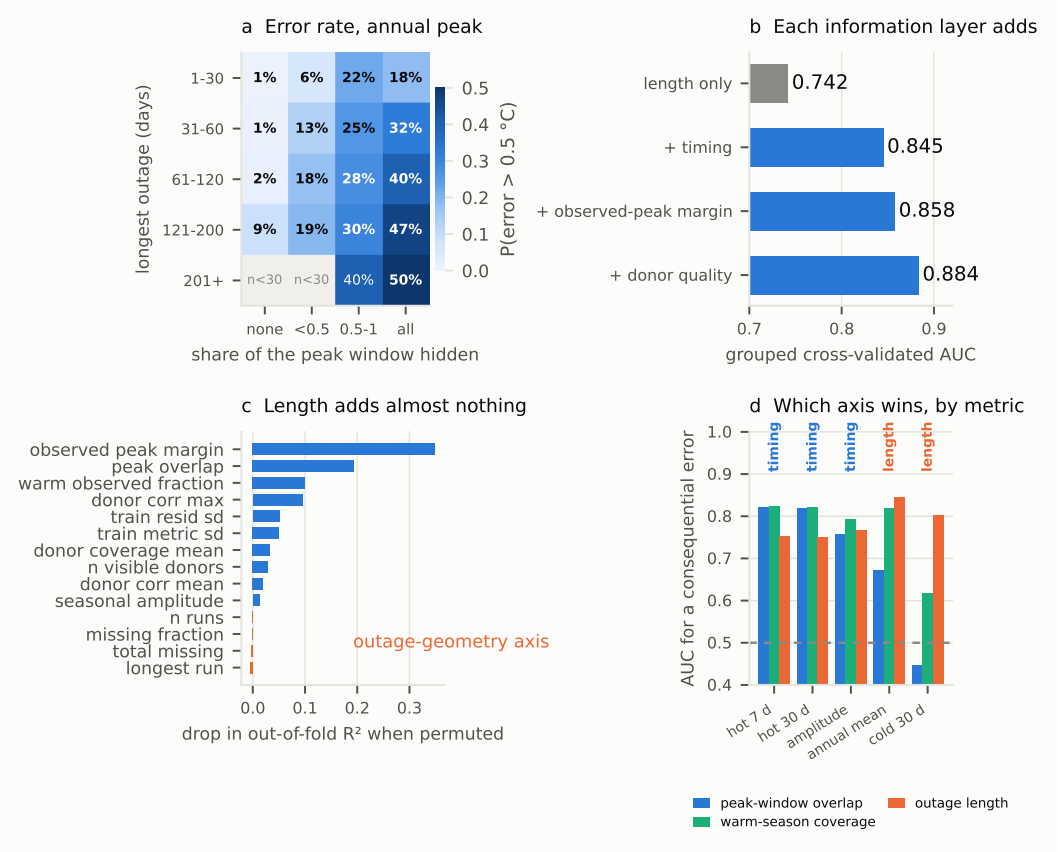


Figure 6. Why position dominates: hidden support, not outage length, carries the error. Development cases (n = 40 704 cases, 110 networks); error bars are network-clustered 95% intervals unless a panel says otherwise. (a) Growth of consequential error along each axis with the other averaged out, computed in bins of both properties: the hottest 30-day mean grows 9.6× along peak-window overlap and 3.5× along missing days, while the annual mean reverses the two (1.5× and 10.2×). (b) Out-of-fold AUC of nested feature sets grouped by network, from outage length alone (0.742) through support (0.845) and observed-peak margin (0.858) to donor quality (0.884). (c) Permutation importance of the decision-time features; once the support is in the model, outage length adds nothing measurable (−0.004 against 0.194 for peak-window overlap). (d) The winning axis by endpoint, the reversal of the main figure computed on the same cases. Source data: `results/v17_fitness/step5_mechanism/` and `step22_robustness/`.

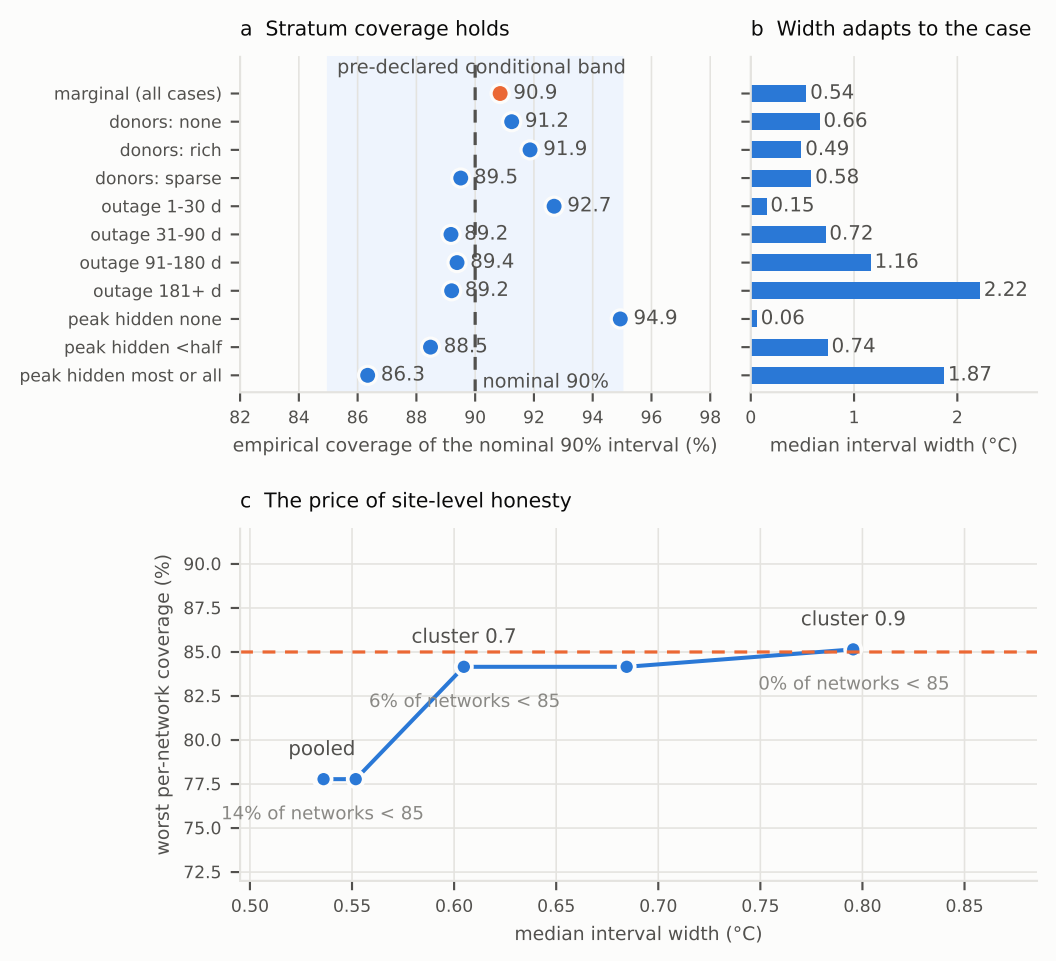


Figure 7. Is the predicted error bound honest enough to rank station-years? (a) Coverage of the nominal 90% interval by donor-regime stratum on the reserved panel: 90.9% marginally, inside the pre-declared 0.87–0.93 band, and 86.3–94.9% within every pre-declared stratum; the dashed line is the nominal level. (b) Interval width in °C against the share of the peak window the gap hides, from 0.06 °C when the window is untouched to 1.87 °C when most of it is hidden, with network-clustered bands, so the bound tracks the quantity it is meant to bound. (c) Width price of a site-level rather than pooled guarantee: the site-level quantiles reach no network below 0.85 at a 1.5-fold widening, but that level was chosen on the opened reserved panel and stays exploratory. Development cases unless a panel says otherwise. Source data: `results/v17_fitness/step8_conformal/` and `amendments/e6_support_width_uncertainty/`.

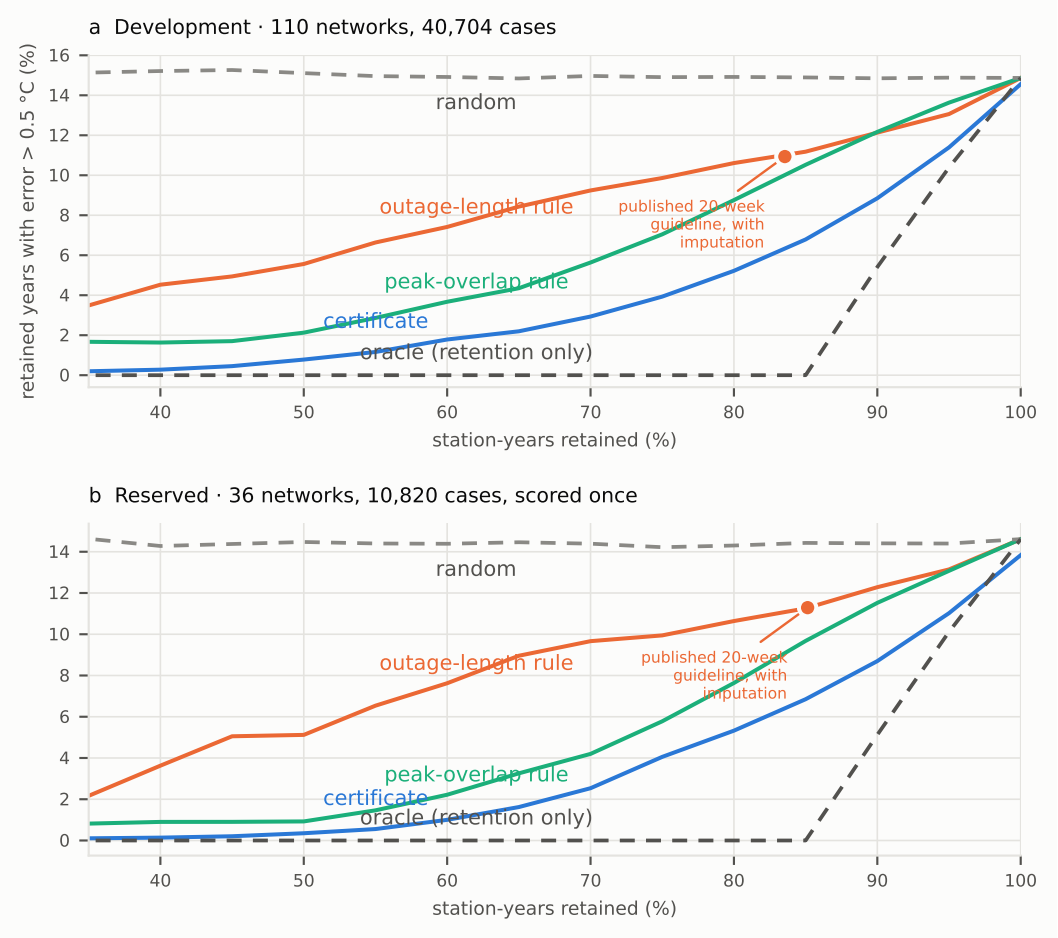


Figure 8. What equal retention costs under each admission rule. (a) Development cases (n = 40 704 cases, 110 networks): the share of retained station-years whose hottest 30-day mean is wrong by more than 0.5 °C, against the share of station-years retained, for random retention (grey), the gap-length rule (orange), peak-window overlap alone (blue) and the calibrated screen (black). (b) The same curves on the 36-network reserved panel (10 820 cases), which was opened once after the configuration was frozen. Curves are network-clustered means with 95% bands; the square marks the operating point quoted in the text, 70% retention, where the screen leaves 2.5% against 9.7% for the length rule, a contrast of −7.13 points ([−8.97, −5.36]). Source data: `results/v17_fitness/step4_ceiling/` and `step10_confirmation/`.

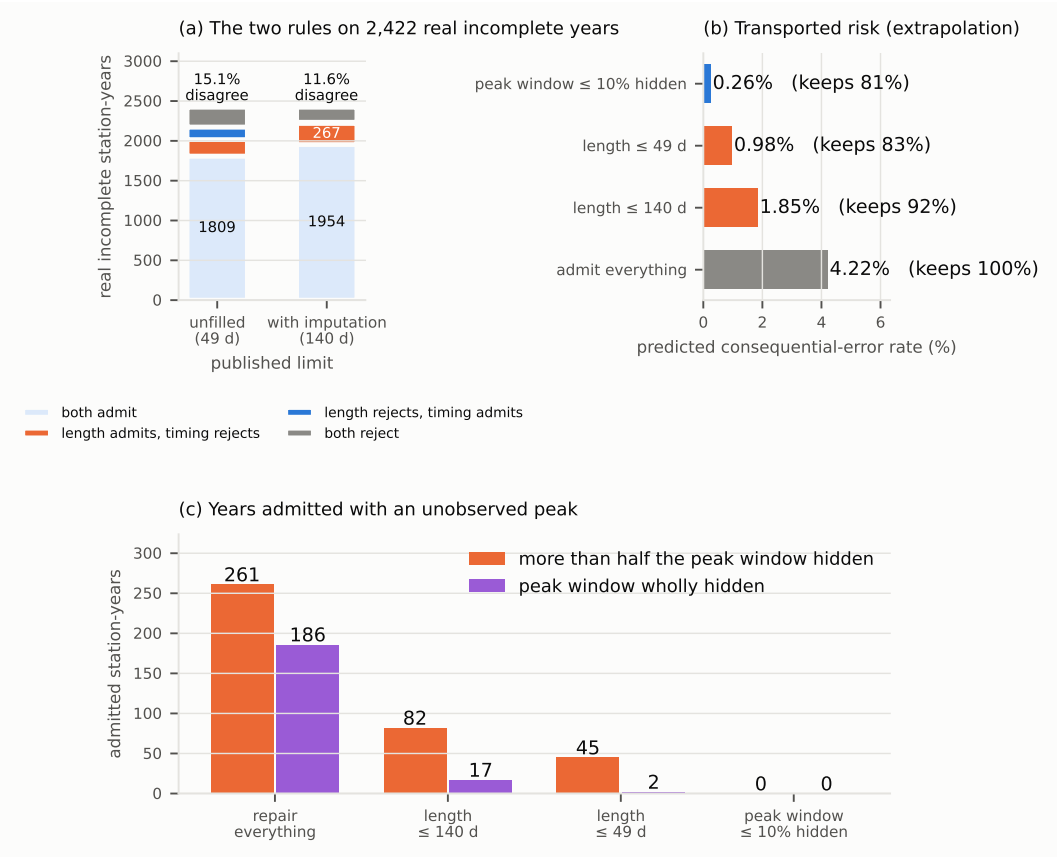


Figure 9. What the two admission rules do to records that really have gaps. (a) The length rule and the support screen applied to the 2422 incomplete station-years of this archive that have never been decided (of 7536 station-years in total, 5114 of them complete): they disagree on 15.1% of admissions ($[13.4, 16.8]$%, 136 network clusters), and on 8.5% the length rule admits a year whose climatological peak window is more than half hidden. (b) Predicted, not observed, consequential error for each admitted set (there is no truth in these years); the screen's admitted set is the more conservative one, and the prediction is miscalibrated by at most 6.2 points in its worst decile. (c) Station-years the length rule admits with more than half of the peak window hidden (n = 261); 28.7% of the 2422 years hide at least one day of the climatological peak window (median 9 missing days, 90th percentile 128). "Peak window" here is the climatological window from the station's earlier years, a proxy: the positive predictive value of that proxy for a wholly unobserved realised peak is 0.74 on designed cases. Source data: `results/v18_closure/e2_archive_deployment/`.

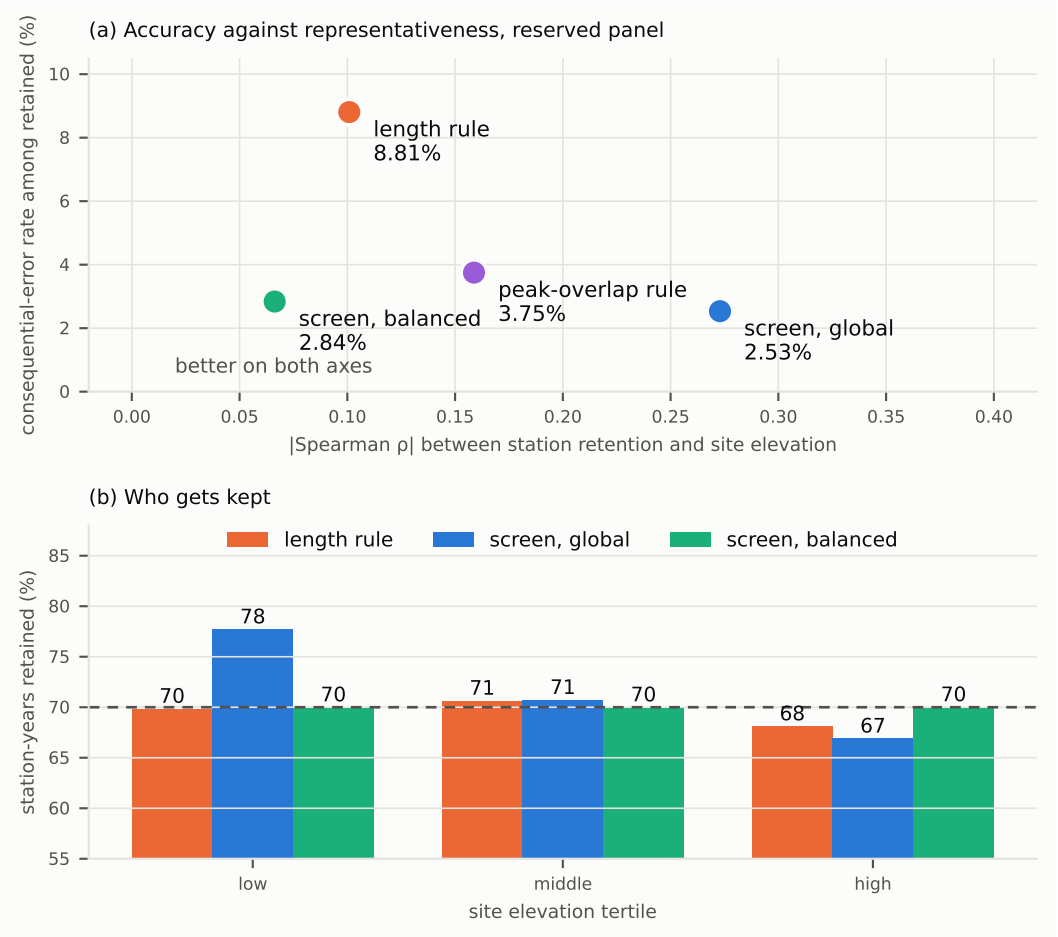


Figure 10. Which station-years the screen costs, and what balancing them would cost. (a) Across the development networks, the higher the correlation between a station's retention and its site elevation, the larger the consequential-error rate under the calibrated screen ($\rho = -0.259$, $p = 0.011$, 43 stations with GAGES-II attributes); the screen appears to select on difficulty, which tracks catchment type. (b) On the reserved panel, retention by elevation tertile under the length rule, the calibrated screen and the screen with thresholds fitted within strata: the screen retains 10.8 points fewer high-elevation station-years than low-elevation ones ($[-15.4, -0.3]$, 36 network clusters), and balancing on elevation recovers that at a cost of 0.31 points of consequential error. Bars are network-clustered 95% intervals. Source data: `results/v18_closure/e4_representativeness/` and `amendments/faithful_baseline/`.